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Development of a multimodal strategy to investigate speech processing mechanisms in infants

Silvia Valls Santafé

Universitat Pompeu Fabra
Barcelona

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Thesis supervisor(s):

Dr. Núria Sebastián Gallés,

(Speech Acquisition & Perception Group, Center for Brain and Cognition)

Ibtissam Ghailan Tribak

(Speech Acquisition & Perception Group, Center for Brain and Cognition & Institut de Ciències Fotòniques)

Universitat Pompeu Fabra
Barcelona

Dedicatory

I dedicate my Bachelor's thesis to my grandfather, who has been my greatest supporter throughout my Biomedical Engineering degree. From the very first day, your presence has given me the strength to keep going, even in the most difficult moments.

You have celebrated every step with me, listened with patience and reminded me of my worth when I doubted myself. Your quiet encouragement, your pride in everything I do and the love you show in a thousand small ways have meant more than I could ever put into words.

This thesis is not just a personal milestone, it is a tribute to everything you have done for me.

Thank you for being my constant, my guide and my inspiration.

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Abstract

Understanding how infants process speech signal is a question of interest in neuroscience, because it provides insight into the early language acquisition and cognition development. However, investigating these mechanisms requires neuroimaging tools capable of capturing brain activity, adapted to the specific challenges of measuring this population. This Bachelor's thesis contributes to a broader project that aims to uncover speech processing mechanisms in 4-5-month-old infants, presenting the development of a multimodal strategy that integrates electroencephalography (EEG), functional Diffusion Correlation Spectroscopy (fDCS) and functional Near-Infrared Spectroscopy (fNIRS) to acquire data by characterizing the neuronal activity and its relation to oxygen metabolism (i.e., energetic demand and oxygen consumption). To do so, this project aims to develop a multimodal system that combines optical techniques (fNIRS and fDCS) with EEG. Optical techniques would provide hemodynamic response, while EEG will quantify the electrical activity of the neurons enabling acquisition with both high temporal and spatial resolution. To implement this approach, a custom pipeline combining Python with 3D modelling was developed to automate the generation of sensor configuration (EEG electrodes, fNIRS and fDCS optodes) while respecting anatomical and technical constraints. Moreover, it also addresses the challenge of reducing the attrition rate caused by poor signal quality due to hair related issues in fDCS measurements. To address this challenge, a novel mechanism was developed based on pressure adjustment. Following multiple iterations, some models were prototyped through 3D printing and CNC (Computer Numerical Control) turning to ensure mechanical robustness and usability. The final system has the potential to improve contact across diverse hair types, to reduce setup time and to minimize participant exclusion. While further improvements are needed before deployment, the system represents a meaningful step toward enabling simultaneous acquisition of EEG, fNIRS and fDCS data in early infancy.

Keywords

Multimodality, infant-compatible system, neuroimaging, optical engineering, signal quality, EEG, fNIRS, fDCS, 3D printing, language discrimination.

Author Declaration

The Author, Silvia Valls Santafé, discloses that the writing of this document, *Development of a multimodal strategy to investigate speech processing mechanisms in infants*, involved the use of AI, particularly ChatGPT for the purpose of language review and improvement of clarity. The Author warrants, however, that the contents of this work are a genuine human creation, as the results obtained from AI have been duly reviewed and edited by Silvia Valls Santafé.

Prologue

Human language is one of the most complex capabilities of our species, and its acquisition begins remarkably early, before infants even speak a single word. In a world full of noise and unpredictability, their brains start to extract meaningful patterns from speech, laying the neural foundations for communication. This process, which unfolds without prior linguistic knowledge or structured guidance, is both inspiring and intellectually captivating. It highlights the innate human capacity to learn from the environment and represents the essence of how we begin to make sense of the world. Studying this phenomenon not only bridges neuroscience and linguistics but also reflects the core of human development.

I have always been deeply interested in machines, their precision, complexity and the way they bring abstract concepts to life. Neuroimaging technologies such as fNIRS and fDCS captivated me not only for their role in studying the brain, but also for the advanced physical principles that make them work. Seeing these systems up close was a unique opportunity to merge my passion for engineering with scientific discovery. As a future engineer who enjoys understanding how things function at a fundamental level, learning the physics behind these devices was the perfect opportunity to do this experience especially rewarding.

Turning that curiosity into scientific research quickly encounters a technical reality: measuring brain activity in infants is extremely challenging due to their constant movement, limited attention span and anatomical constraints. Infants get easily bored, need to stay close to their parents, and cannot follow instructions, making it difficult to obtain clean and reliable signals in a short time window.

Rather than accepting those limitations, this Bachelor's Thesis approached them as an engineering challenge. Could a system be designed that is gentle, fast, inclusive and precise enough to capture brain activity? This question led to a multidisciplinary effort that merged biomedical engineering, computer programming, prototype design and a great deal of practical trial and error.

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1. INTRODUCTION

1.1 Foundations for infant language research

Language can be considered as a code shared by a group of people that serves for communication purposes. Understanding the way in which infant's brain processes language during the earliest stages of development is crucial to understand the foundations of language acquisition in infants, defined as individuals between 3 and 12 months of age, who have no prior linguistic knowledge [1]. Language is one of the most complex cognitive skills humans develop, and it starts taking shape earlier than previously assumed. The ability to discriminate between languages during the first months of life provides valuable insight into the mechanisms underlying early perceptual narrowing in language processing [2].

This early ability to discriminate between languages offers a window into how the brain begins to specialize in language [3], even before infants start speaking. To investigate this process, it is important to use neuroimaging techniques that can capture how the brain reacts to different types of linguistic input over time.

However, studying how infant brain process linguistic input, requires not only suitable neuroimaging techniques but also methodological approaches adapted to the specific challenges of working with this population. To begin with, infants have extremely short attention spans and can easily become uncomfortable or lose interest. If a stimulus or device element causes a minor discomfort (e.g., having a cap place on their head) the infant may start crying. Consequently, every step must be complemented rapidly. Therefore, it is essential to design systems that are fast to position, comfortable, cheap and secure enough to stay in position, even if the infant moves.

Additionally, because infants must remain with the parents, the device must be as small, lightweight and portable as possible, allowing the recordings to take place while the infant is seated on the caregiver’s lap. This portability is essential not only for maintaining the infant’s comfort and sense of security, but also for enabling data collection in a naturalistic environment without requiring rigid positioning or separation from the parents. Accordingly, obtaining neural recordings in awake infants demands equipment and protocols specifically tailored to the distinctive behavioural profile of this population.

In addition, infant neuroimaging is further complicated by biological factors. Neurodevelopment is a dynamic regulated process, influenced by a complex interplay of genetic, environmental and physiological factors. As a result, infants undergo constant and rapid changes [5], not only in size and brain development but also in the physical properties of their brain tissue [6]. These changes make difficult to design a reliable neuroimaging system that works equally well across different developmental stages.

Although the infant brain develops rapidly, it will reach maturity several months after birth. Sensory areas mature early and rapidly, while associative regions, such as the frontal cortex, develop more gradually and continue into adolescence [7]. In the first months, for example, myelination is still minimal, which limits the use of certain imaging techniques [7]. For instance, Magnetic Resonance Imaging (MRI) relies on fat presence in the tissue, whereas infant brain contains a large proportion of water [7]. Moreover, MRI is extremely sensitive to motion artifacts and usually requires infants to be asleep or sedated [7], which is not ethically questionable, especially in studies where the infant needs to be awake and engaged. Brain activity under these conditions may become distorted.

Given these constraints, optical neuroimaging techniques have emerged as valuable alternatives. They are non-invasive, silent, relatively cost-effective, more affordable than MRI or MEG (magnetoencephalography recording), and portable [4], an essential feature when studying infants, who move frequently and cannot be easily constrained in standard setups. Nonetheless, one challenge observed in a previous study [Ghailan et. al., in preparation] in the laboratory where this thesis is conducted, is that more than 50% of the excluded data was due to a non-optimal optical contact with the scalp due to the interference of the hair that was leading to a significant decrease of the total amount of photons per second that was been measured compromising the quality of the data.

1.2 Neuroimaging techniques

According to the neurovascular principle, the microvasculature supports neuronal activity by supplying oxygen and nutrients required for cellular respiration, the biochemical process by which cells generate energy for their functions [8]. Consequently, activation of the neuronal compartment increases the demand for oxygen, which is met by the arteries delivering higher concentrations of oxygenated blood to the local areas through increased blood flow (BF) [9]. As a result, changes in the vascular compartment can be used as an indirect indicator of triggered functional activity. However, this measurement is characterized by a slower response time than that of electrical activity [9].

1.2.1 Electroencephalography (EEG)

EEG is a non-invasive technique that records electrical activity produced by the neurons through electrodes placed on the scalp [4]. The signal primarily reflects the summation of slow postsynaptic potentials generated by large populations of synchronously active neurons [4]. One of the most common applications of EEG is the study of neural oscillations, which are rhythmic patterns of electrical activity generated by large populations of synchronously active neurons. These oscillations can be measured as changes in voltage over time and may reflect different cognitive processes or the brain's synchronization to external stimuli [4].

1.2.2 Optical imaging techniques

Functional Diffusion Correlation Spectroscopy (fDCS) and functional Near-Infrared Spectroscopy (fNIRS) are optical techniques founded on the same physical principle of light-tissue interaction [9]. Near-infrared light is not significantly absorbed by the main chromophores in the tissue (i.e., water, fat, collagen and haemoglobin); rather, it scatters when it interacts with molecules in the tissue, particularly by moving particles such as blood cells [9]. This scattering enables NIR light to penetrate deep into the tissue and extract physiological information, such as the concentration of oxy- and deoxyhaemoglobin or BF [9], using fNIRS or fDCS, respectively.

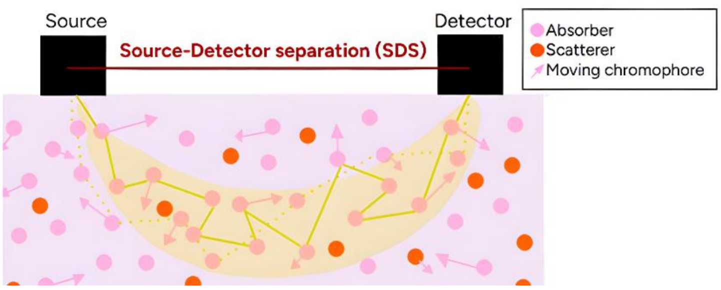

Figure 1. Light propagation path in optical neuroimaging. Photons emitted by the source diffuse through the tissue, following the banana-shaped trajectory (yellow band) that represents the most probabilistic path of the detected photons. The red line indicates the source-detector separation (SDS), whose distance determines the depth of tissue sampled.

fNIRS measures the concentration of oxyhaemoglobin (HBO_2) and deoxyhaemoglobin (HB) by analysing the attenuation of NIR light as it passes through biological tissue [9]. This is achieved by comparing the intensity of the emitted light (I_0) with that of the detected light (I), applying the logarithmic transformation to quantify the changes:

$$A(t) = \log \left(\frac{I_0}{I(t)}\right)$$

where A(t) is the attenuation (or optical density) at time (t). This attenuation is related to the concentrations of chromophores via the modified Beer–Lambert law, which accounts for scattering effects in tissue. By using multiple wavelengths, fNIRS exploits the different absorption spectra of HBO_2 and HbR, allowing their contributions to be differentiated [9].

The other optical technique is fDCS, which is an optical technique that injects long coherent NIR light of a wavelength of 785 nm into human tissue and some of the light is recollected by the detector at a certain distance from the source of light. The diffusion of a photon in human tissue occurs when the direction of photon's propagation gets randomizes after some scattering events which is when the diffusion effect takes place [9]. The photons scatters when interfering with molecules from the tissue. Some of the diffusive light that has interacted with the tissue molecules through mainly scattering and absorption, will be detected by a photo-count detector. Phase variations in long coherent light give rise to constructive and destructive interference, resulting in the formation of dynamic speckle patterns [9]. The light detected from a speckle encodes information about the dynamics of the moving scatterers (i.e. red cells) [8] [9]. To quantify the temporal fluctuations in the detected intensity I(t), fDCS computes the normalized intensity autocorrelation function (g_2):

$$g_2(\tau) = \frac{\langle I(t) \cdot I(t+\tau)\rangle}{\langle I(t)\rangle^2}$$

where τ is the delay time and ⟨·⟩ denotes temporal averaging. This function is then transformed into the electric field autocorrelation function $g_1(\tau)$ by applying the Siegert relation, where β is an instrument related parameter. The β-value is derived from the inverse of the number of detected modes. β-value is 0.5.

$$g_2(\tau) = 1 + \beta|g_1(\tau)|^2$$

The semi-infinite solution for $g_1(\tau)$ follows an exponential decay, which depends on the motion of red blood cells. This decay contains the parameter αD_b, proportional to BF:

$$g_1(\tau) \sim e^{-\sqrt{\alpha D_b}\cdot \tau}$$

In this context, α denotes the ratio of scatters in motion relative to the total number of scatters, while D_b is the effective diffusion coefficient of the moving scatterers [9]. Finally, the Blood Flow Index (BFI), serves as a quantitative measure of cerebral blood flow:

$$BFI = \alpha D_b$$

fDCS provides localized measurements of cerebral blood flow with a spatial resolution of approximately 1-3 cm, depending on optode placement precision [8]. This technique has only been used in this field of research in 2 studies conducted by the Speech Acquisition and Perception (SAP) group (UPF) in collaboration with the Medical Optics group (*Institut de Ciències de Fotòniques* - ICFO).

The concentrations of HBO_2 and HbR measured in fNIRS, reflect local changes in oxygenation associated with neural activity. These signals, when combined with fDCS data, allow for the estimation of the cerebral metabolic rate of oxygen consumption ($CMRO_2$), an indicator of the brain's energy use. This is relevant because combining both techniques provides a more complete picture of neurovascular function, capturing both oxygen consumption and cerebral perfusion.

1.3 State of the art

1.3.1 Neural and cognitive foundations of early language acquisition

The early acquisition of language is a complex and highly dynamic process that involves multiple cognitive and neural mechanisms. From continuous streams of speech, infants must gradually detect and learn the linguistic patterns [1]. Advances in neuroscience have provided significant insights into how the infant brain processes speech, using non-invasive neuroimaging techniques, mainly EEG and fNIRS [10].

In this context, research suggests that newborns exhibit early neural sensitivity to speech, as they can differentiate between different languages and dialects during the first days of life [11]. fNIRS studies have shown that neonates engage both hemispheres when processing native and foreign dialects, with a transient right-hemisphere shift observed when discriminating native from flattened dialects. These findings highlight the importance of prosodic cues in the initial stages of language perception [11]. Similarly, EEG studies have shown that bilingual and monolingual infants develop different neural strategies for processing linguistic input, with bilinguals exhibiting a more distributed pattern of activation across both hemispheres [3].

The explain these early language sensitivities, the Time and Intensity Grid Representation (TIGRE) model suggests that language discrimination in infancy is guided by prosodic features when distinguishing between languages. This model holds that, at birth, infants categorize languages based on rhythmic properties before acquiring more detailed phonetic discrimination abilities. Over time, as infants gain more exposure to linguistic input, their processing mechanisms become more refined, integrating additional speech features to support language differentiation [3].

Consistent with TIGRE, behavioural and neuroimaging studies have shown that infants rely on rhythmic, statistical and phonological cues to segment speech. Prosodic features including variations in pitch, tempo, stress and intonation, play a crucial role in helping infants detect word boundaries [1]. Research indicates that both monolingual and bilingual infants primarily rely on prosodic and rhythmic cues to segment speech. However, they may differ in how they integrate these cues, possibly as an adaptation to their linguistic environments, although this remains an open question [3]. Neuroimaging studies have further supported these findings, showing distinct brain responses linked to these segmentation mechanisms [12].

Moreover, bilingualism provides an opportunity to examine neuroplasticity in early language development. Studies using fNIRS and EEG revealed specific patterns of brain activity in bilingual infants that differ from those seen in monolinguals [12]. Whereas monolingual infants typically show left-lateralized responses to speech, bilinguals engage broader neural circuits across both hemispheres [12]. This suggests that exposure to multiple languages from an early age induces functional and structural changes in the developing brain, enhancing cognitive flexibility and auditory processing abilities.

1.3.2 Advances in infant neuroimaging for language research

The study of early language acquisition has benefited from non-invasive neuroimaging techniques, which allow researchers to examine the developing brain in real time. fNIRS measures hemodynamic responses related to speech processing and has been widely used to investigate infants' responses to linguistic stimuli, including accent discrimination and bilingual exposure effects [13] [12]. Recently, diffuse correlation spectroscopy (DCS) has emerged as a promising tool for measuring cerebral blood flow at the microvascular level, providing additional information about cerebral hemodynamic activity [8].

Functional near-infrared spectroscopy (fNIRS) has proven to be an effective and non-invasive tool for investigating the neural basis of speech processing in infancy [12]. Studies using this technique have revealed early functional specialization in regions such as the superior temporal and inferior frontal gyri, especially in monolingual infants exposed to familiar languages [12]. Furthermore, research has shown that bilingual infants may follow distinct developmental trajectories, possibly reflecting increased demands on neural systems responsible for managing dual linguistic input [12].

In addition to these findings, previous research [13] demonstrated that fNIRS is sensitive enough to detect cortical responses to phonetic differences, such as accent contrasts, in 5-month-old infants [13]. This work emphasizes the method's capacity to capture subtle differences in brain responses, even in pre-verbal populations, extending its applicability beyond broad linguistic categories to more specific aspects of speech perception. These findings highlight the sensitivity of fNIRS to differences in language experience and reinforce its value for longitudinal and comparative studies in early language acquisition [12][13].

Recent research has demonstrated the feasibility of using fDCS to detect subtle changes in cerebral blood flow (CBF) in response to somatosensory stimulation in neonates [14]. In a study involving preterm infants, a soft optical probe with a single source and multiple detectors was positioned over the hemisphere contralateral to the stimulated hand, allowing for targeted measurement of regional hemodynamic activity [14]. The study successfully identified localized increases in CBF, highlighting the sensitivity of fDCS to functionally relevant neurovascular responses, even at very early developmental stages [14]. Although the temporal resolution of fDCS is limited compared to EEG, this technique offers direct access to microvascular dynamics. fDCS and fNIRS provide complementary information, since both measure local changes in oxygenation related to neuronal activity. Specifically, increases in cerebral blood flow led to higher levels of HbO_2 and lower levels of HbR, enabling indirect assessment of neural activation [14][4].

While studies have been conducted in preterm and neonatal populations, there is a lack of evidence in young infants (aged 3-12 months). To address this, 2 studies were conducted in the *Laboratori de Neurociència* at *Universitat Pompeu Fabra* to validate the use of fDCS in 4–5-month-old infants. The first study assessed whether fDCS could detect changes in CBF evoked by visual cognitive processing in the occipital lobe and aimed to identify a protocol that would allow the comparison of CBF and electrical activity measurements, considering the differences in temporal resolution between the techniques [Ghailan et al., in preparation]. The second study tested the sensitivity of fDCS to auditory processing, specifically to words structures, targeting the frontal and temporal regions [Ghailan et al., in preparation].

1.4 Objectives

Measuring brain activity in awake infants is a challenging issue as explained in the previous section. Most research uses either EEG, with excellent temporal resolution (ms) but very limited spatial resolution, or fNIRS, with very good spatial resolution but very limited temporal resolution. Therefore, the objective of this Bachelor's Thesis (BT) is to contribute to the development of a multimodal neuroimaging system for studying early language processing in 4–5-month-old infants with high temporal and spatial resolution by (i) improving data usability through reanalysis of existing recordings, (ii) designing a mechanism to improve scalp-optode contact in fDCS measurements, and (iii) designing and developing a sensor distribution that integrates EEG, fNIRS and fDCS optodes while respecting anatomical and technical constraints.

To achieve the overall objective of this BT, several specific goals were pursued: (1) to develop skills in the preprocessing and analysis of neurophysiological data relevant to infant brain research; (2) to gain experience in neuroscience laboratory procedures and the management of participant-related protocols; (3) to acquire a general understanding of the physical principles behind optical neuroimaging techniques, specifically fDCS, fNIRS and EEG; (4) to design a multimodal head-mounted setup through 3D modelling and 3D printing; and (5) to design, print and prototype a mechanical mechanism capable of applying controlled pressure to the optodes of fDCS in order to ensure stable and reliable contact with the scalp, particularly in the presence of hair-related interference.

To provide a visual overview of the approach adopted to achieve the objectives, *Figure 2* illustrates the methodological framework followed throughout the BT.

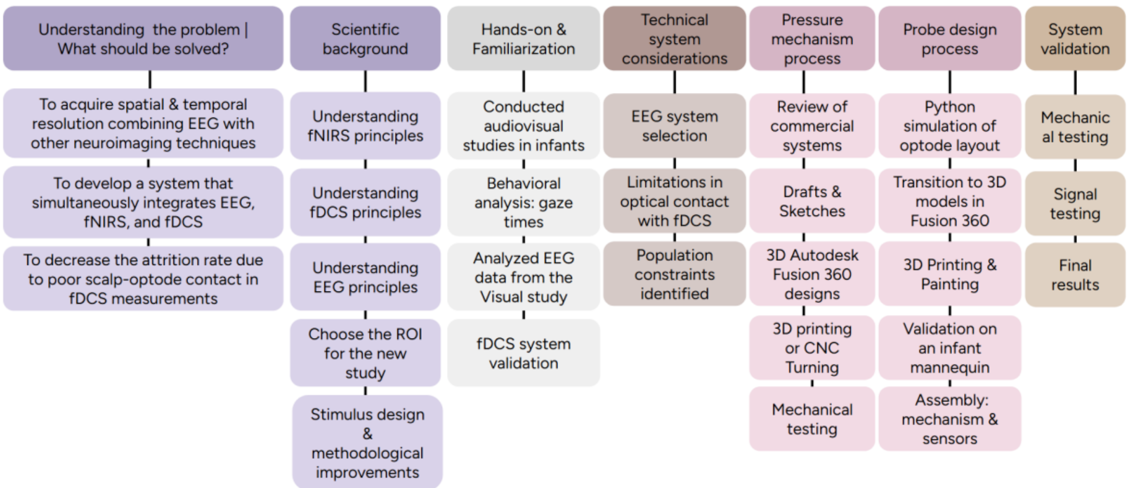

Figure 2. Diagram summarizing the main stages followed throughout the project. Each step builds toward the development of a multimodal neuroimaging setup. The top row presents the main phases of each stage, while the underlying boxes correspond to the specific subtasks required to achieve them.

2. MATERIALS AND METHODS

2.1 Development of the multimodal study experimental protocol

The integration of fDCS and fNIRS aims to adress the limitation of the original study [3] which only provided temporal resolution, by also enabling spatial resolution.

The region of interest (ROI) of the present study comprises both temporal lobes, with particular emphasis on the left perisylvian region, an area involved in multiple components of speech and language processing. This includes Broca's and Wernicke's areas, which are essential for speech production and comprehension [16]. The temporal region was targeted due to its role in auditory processing, particularly in infancy. This selection was grounded in previous literature, including work by [13], as well as [12], which studied these areas and provided evidence that neural responses in these areas are already functionally specialized during infancy. Moreover, the occipital region was selected as a control area because it is not directly involved in language processing in infants with typical development [17]. This choice was to ensure that any effects observed in language related areas are specific to language processing and not due to general neural activity or measurement noise. All targeted regions are illustrated in *Figure 20*.

A detailed explanation of the stimulus design and methodological improvements is provided in *SI - 1.1 Stimulus and workflow design.*

2.2 Development of expertise in infant studies

To ensure an understanding of infant studies, several experiences were conducted.

2.2.1 Auditory study

The first experience involved participation in an auditory study with 6-month-old infants at the *Laboratori de Neurociència, Universitat Pompeu Fabra*, which included guiding families through the procedure, obtaining informed consent and observing infants' responses to auditory stimuli. Although this age group was slightly older than the target population of the current project (4–5 months), the experience allowed to observe infant behaviour and experimental procedures. This process provided practical insights into the protocol's structure, the importance of optimizing preparation time and the need for engaging stimuli to maintain infant attention. These observations were applied in the design of the new setup, ensuring that the system could be rapidly installed while maintaining infant comfort and attention.

2.2.2 Visual study

The second experience involved a visual study combining EEG and fDCS aimed at validating the reliability of fDCS for measuring brain activity in the occipital lobe. EEG, recorded simultaneously, served as the gold standard for comparison. In the context of this thesis, only the EEG data has been analysed, while the experimental design and data collection were conducted prior to this work.

Forty-two infants aged 4-5 months were recruited. Following a data quality assessment, 15 infants were excluded: 6 due to fussiness and 9 due to poor data quality. Prior to data analysis, 8 more infants were excluded: 6 due to missing video recordings, 3 due to video format errors, 3 due to poor data quality, and 1 because it was impossible to determine which stimulus the infant was looking at because the mirror was poorly positioned and the changes in luminance of each stimulus were indistinguishable in the video recording.

In this thesis, the focus was on the first phase of the experimental protocol, as presented in *Figure SI- 1*, out of the 3 that included the study. This phase consisted in presenting 3 times a sequence of three stimuli. Each of stimuli had a different role, detailed as follows: (i) the baseline to establish a stable neural and vascular baseline, (ii) the attention getter to attract infants' attention to the screen before the main stimuli and (iii) the main stimuli that consisted of a checkerboard (CK) pattern flickering at 5 Hz.

2.2.2.1 Analysis of data

Analysing infant behaviour during visual stimulation was crucial for identifying limitations of the setup during measurement that influenced the present study, where the multimodal setup was optimised to minimise data loss. As this study involved EEG and fDCS, two out of the three techniques used in the current study, it allowed for an initial assessment of their performance.

This analysis started by determining the amount of time that each participant looked to the screen during the stimuli presentation. This process was completed using video recordings captured by a Logitech camera placed above the Asus screen. A minimum of 50% looking time was required for data inclusion. The camera view included a mirror behind the setup, showing the screen from the infant’s perspective.

As specified, the analysis focused on Phase 1, specifically on the CK stimulus. Gaze onsets and offsets were annotated in Excel. Due to the camera angle, apparent gaze direction was sometimes misleading. Therefore, only gazes lasting at least 2 consecutive seconds during CK were considered valid. The mirror helped verify screen content, but its occasional misalignment posed a limitation. As an alternative, stimulus detection was supported by analysing luminescence variations, CK being brighter than attention getter. Stimulus onset times were logged in Excel to verify timing accuracy. The annotations were validated with another researcher to ensure consistency and reliability.

Following the video analysis, only those with sufficient gaze duration were selected for subsequent EEG processing using the APICE pipeline [18] implemented in EEGLAB. Band pass filters to remove low and high frequencies [0.1-40 Hz] and motion artefacts (i.e., eye blinking) correction algorithms were applied to improve the data quality. Bad times and epochs were identified and removed while bad channels were interpolated using the APICE automatic algorithm. After the data was pre-processed, the data corresponding to the CK was epoched. The epoch lasted the whole duration of the CK (8s). A Hanning window was applied to each epoch to decrease the spectral leakage caused by discontinuities. After, the Fast-Fourier Transform was calculated followed by an assessment of the Steady State Visual Evoked Potential (SSVEP) amplitude. SSVEP is defined as the ratio between the peak at the frequency of interest (i.e., 5Hz) and the expected amplitude if no activity at that frequency (5Hz) was present in the stimuli. The expected amplitude can be extracted as the mean value from the amplitude of the adjacent frequencies (i.e., 4.5Hz - 5.5) [18].

2.2.3 fDCS system validation

Although fDCS data analysis from the visual study was not part of this thesis, hands-on experience with the system was fundamental to understand its operation, signal characteristics and sensitivity to different conditions. Therefore, a preliminary validation of the system was conducted. Optical power was measured both at the laser output and after transmission through 10 meters of fibre using a power meter, quantifying signal attenuation caused by the fibre.

Further validation involved heart rate measurement in an adult participant. The right and left probes were placed on each arm (a well-perfused site), secured with bandages. Probes included 785 nm light sources and photon detectors with a 1.5 cm SDS.

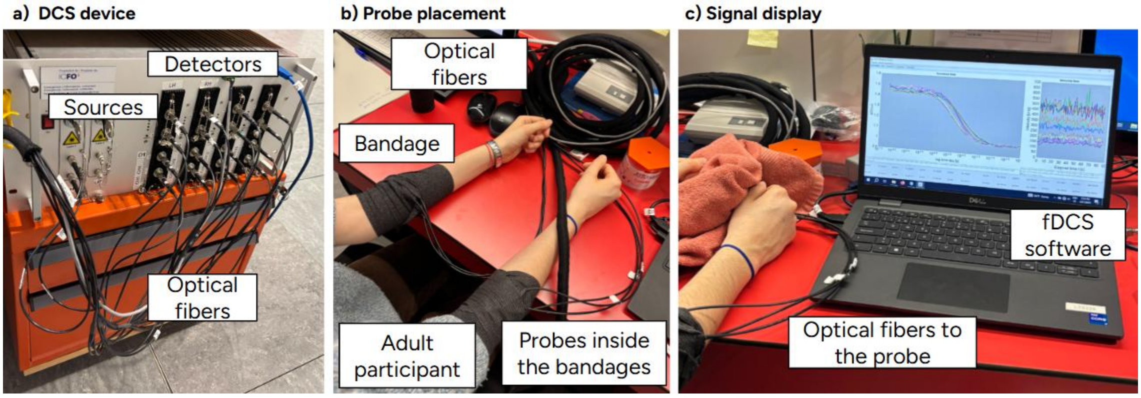

Figure 3. Validation of the fDCS system. The left panel (a) shows the fDCS device setup with fibres and connectors. The middle panel (b) displays the probe placement on the participant's forearms during the test. The right panel (c) shows the system interface with a real-time signal trace on the screen.

Working with fDCS allowed me to become familiar with the technique, including the device setup, the probe placement, fibre positioning, software interface, g_2 curves and light intensity measurements.

Once the familiarization phase was completed, the project moved into the initial design phase of a head-mounted system integrating fDCS, fNIRS and EEG modalities to monitor brain activity in young infants. In addition, this work specifically addresses one of the main limitations in optical neuroimaging, that consists of the degradation of signal quality due to hair interference. The development followed a structured process: first selecting a suitable EEG system, then designing a mechanical pressure mechanism to improve fDCS signal quality and finally integrating these components into fully printed multimodal probes optimized for infant anatomy.

2.3 Technical specifications of the EEG System

An important aspect of the integration process was selecting a suitable EEG system. Although a gel-based cap was considered, it was dismissed due to long setup times, movement-related detachment, limited time for gel application and ethical concerns about scalp preparation in infants, as scraping the infant's scalp to improve contact is not allowed. As observed in the Neuroscience Laboratory (section *2.2.1 Auditory* study), minimising preparation time is crucial to maximise usable data.

A high density commercial wet EEG system with 128 electrodes arranged, according to an extended 10-10 montage, mounted on a silicone mesh cap was chosen instead, offering a faster, safer and more compatible solution for multimodal recordings. To enhance signal acquisition without gel, a mix of water and shampoo is applied to the electrodes beforehand. This humidifies the hair, reducing the impedance at the skin-electrode interface to have a better conductivity, and ensuring the cap is ready before the infant enters the experimental room, thus optimising setup time. Moreover, as shown in the results section *3.1 EEG data analysis from the visual study*, this EEG system proved suitable for simultaneous EEG and fDCS recordings, supporting its selection for the present study.

2.4 Development of a component for enhanced fDCS quality

Hair remains one of the main challenges in infant optical neuroimaging, particularly when using optical techniques. Achieving adequate optical contact between the scalp and the optodes is essential, as dark, dense or thick hair does not allow a proper optical attachment, which substantially avoids photon diffusion into the tissue, degrades signal quality and increases data loss [19].

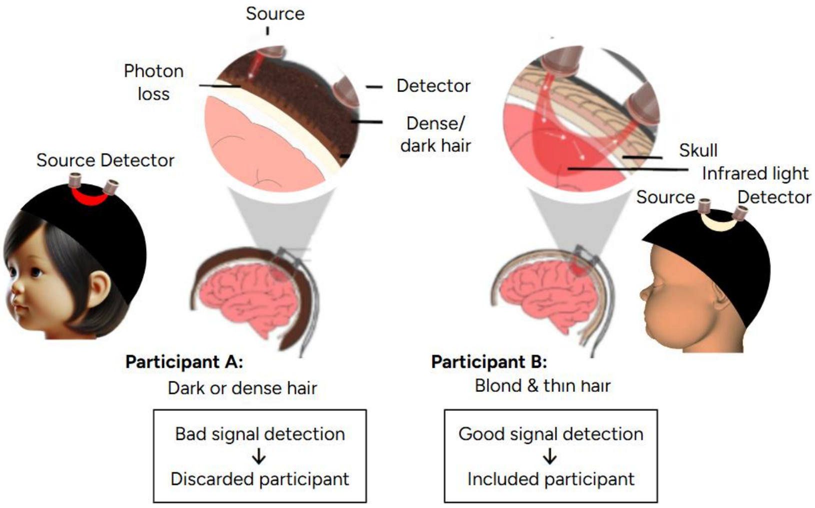

Figure 4. Impact of hair and skin properties on optical signal quality. Illustration created by the author, adapted from [20], showing how dense or dark hair and skin reduce infrared penetration and signal quality.

For instance, a study from the SAP group, [Ghailan et al., in preparation], which combined EEG and fDCS, reported that more than 50% of the excluded infants was due to hair-related issues attributed to poor signal quality caused by dense or dark hair. Although the overall attrition rate was relatively low (33%), this finding introduces a critical sampling bias, favouring infants with lighter or thinner hair and underscores the importance of addressing this limitation in the current thesis.

This finding highlighted the importance of addressing hair related signal interference in fDCS and directly influenced the design of the multimodal probe, prompting the integration of a new component specifically aimed at ensuring stable and optode scalp contact across a wide range of hair types.

2.4.1 Analysis of reference systems for optode fixation

Different commercial solutions have been developed to address the challenge of optode–scalp contact in fNIRS systems for hair interference, but no equivalent approach has yet been implemented for fDCS applications. Existing designs are intended for textile-based fNIRS setups and therefore require alternative strategies when applied to a structurally distinct context. The review of existing fNIRS systems presented below serves as a source of inspiration rather than a direct reference.

The company NIRx, employs spring-loaded grommets, but its limited range of pressure adjustment (there are 4 units per each level pressure) restricts finer adaptation and may result in excessive pressure, raising safety concerns [21]. LUMO, a company focused on smart designs, prioritizes comfort through flexible light guides but proves ineffective in dense hair [22]. CortiVision and Artinis, both commercial providers, rely on manual pressure during setup [23], [24], which can compromise comfort and are not ideal for infant use due to the repeated application of force.

Commercial solutions are designed for specific fNIRS optodes. However, the tip of fDCS optodes differs in shape (see *Figure SI- 4*). This underscores the need for a fixation system specifically tailored to this context, one that ensures reliable scalp contact, minimizes preparation time and maintains infant comfort throughout the procedure.

2.4.2 Comparative evaluation of initial 3D designs

In the absence of existing solutions for fDCS optodes, four initial 3D design concepts were first sketched as architectural drawings and later modelled in Autodesk Fusion 360, each employing distinct mechanical principles for optode fixation and pressure control.

The first design combined a screw and internal gear to adjust the vertical position of a central arm. Rotating the screw moved the optode downward with precision, while a fully enclosed casing ensured infant safety by isolating moving parts. The second design incorporated two screws, horizontal and vertical, allowing independent control of optode positioning and applied pressure. This dual mechanism enhanced adaptability to different head shapes and hair densities. The third concept relied on magnetic attraction. Paired magnets above and below the optode held it in place, enabling pressure adjustment without mechanical parts. The fourth explored a vertical piston system actuated by an adjustable top cover. When pressed, the piston transferred force to the optode, improving contact without exposing mechanical elements, thereby preserving infant comfort and safety. Further details can be found in section *SI -1.2 Conceptual design analysis.*

2.4.3 Exploration of physical prototypes

After discarding the initial 3D designs, explained in section *3.2 Preliminary conceptual analysis of the pressure mechanisms*, the development process shifted toward physical prototyping using simple materials. This approach allowed faster iterations than Autodesk Fusion 360 modelling and enabled real world validation under spatial and mechanical constraints, such as fit on curved infant heads. The prototypes, shown in *Figure SI- 10, Figure SI- 11* and *Figure SI- 13*, were not to scale nor intended for final use, but served to explore conceptual functionality.

The first prototype was a lever-based system housed in a cylindrical casing, which applied pressure through a rod actuated by a spring-loaded locking mechanism. The second used a silicone compression piece with four preset height levels mounted on a rigid base, offering a lightweight and easily integrable solution. The third involved mechanism driven by a threaded screw inside a casing, guiding the optode downward while ensuring stability through a dual-ring base placed above and below the cap. Further details can be found in section *SI - 1.3 Exploration of physical prototypes*. These models were essential in identifying mechanical constraints and refining the final pressure mechanism for safe and effective use in infant neuroimaging.

2.4.4 Optode pressure printed mechanisms

2.4.4.1 Spring-loaded mechanism

The design addressed limitations of the previous prototype through a cylindrical casing integrating a screw and a spring. The optode rested on the spring, enabling passive pressure from the infant's scalp to ensure proper contact. Additionally, a safety ring limited screw descent, maintaining pressure within a safe range. To create an infant-friendly appearance without adding weight, initial animal-shaped pieces (*Figure SI- 15*) were replaced with lightweight animal stickers. Initially designed for a flexible fabric cap, the casing base was later adapted to integrate into the probe once the spatial layout was finalised, ensuring stability while preserving the essential features.

Beyond mechanical integration, the system also enhanced setup efficiency in research settings. When the family would arrive at the laboratory, and one of the researchers would examine the infant's hair, specifically the density, to estimate the necessary pressure for each optode. While one member of the team would explain the study to the parents, another would prepare the setup, pre-adjusting each holder based on this initial evaluation. Once the probes are placed on the infant, signal quality could be immediately checked through the fDCS software. If any adjustment is needed, the pressure would be redefined by simply rotating the screw, without removing or repositioning the holder, making the process faster and more convenient.

The components of the pressure mechanism were printed on a Formlabs 3 stereolithography (SLA) 3D printer. This choice was driven by the need for mechanical precision and personalization, ensuring each part aligns the probe with accuracy and holds it securely. SLA printing with a photopolymer resin fulfils these requirements by building each part layer by layer from a liquid that the laser cures into a rigid solid [26].

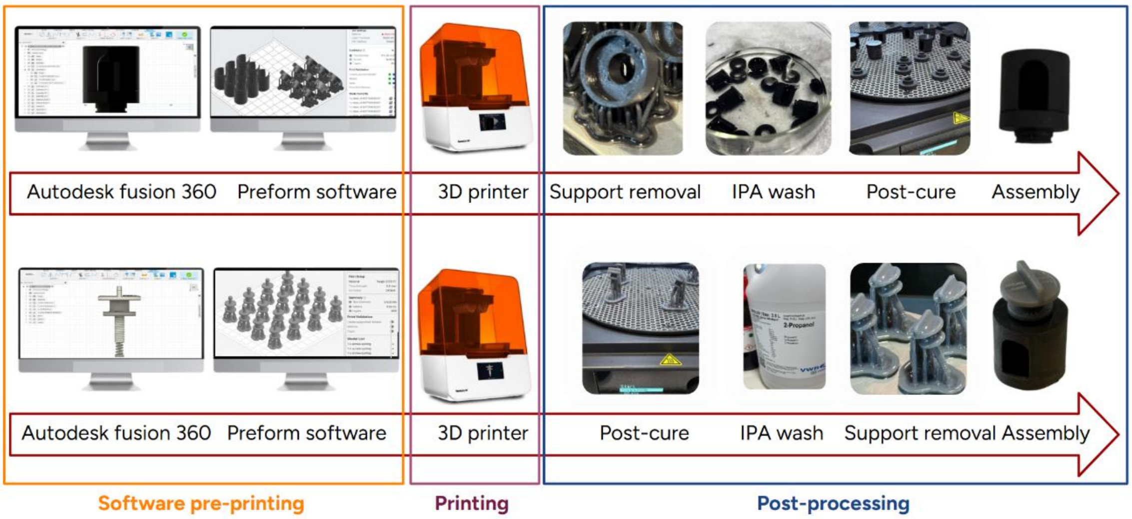

Figure 5. Detailed pipeline for the 3D printing of the optode-pressure mechanism. Top row: bottom-casing process. Bottom row: identical steps but in a modified order to protect the ultra-thin screw-springs.

Formlabs black resin V4 [25]was selected as the photopolymer of choice. Mechanically, cured Black Resin V4 exhibits a tensile strength of 65 MPa [25], enough to absorb the pull and twist forces required when inserting or removing the component from the probe without approaching its elastic limit. It has a tensile modulus of 2.8 GPa with a flexural modulus of 2.2 GPa [25] that keeps the casing essentially rigid under load, allowing the screw allowing to be tightened without the plastic bending or yielding. Moreover, its matte black surface absorbs unwanted light rather than reflecting it, preventing scatter and reflections that could interfere with the sensor, preserving the reliability of each measurement.

In its uncured state Formlabs black resin V4 behaves as a liquid photosensitive material, which, when exposed to a focused ultraviolet (UV) laser, undergoes rapid cross-linking to form a solid polymer network [26]. The minimal layer thickness afforded by SLA ensures that curved surfaces remain smooth, and edges stay sharp. The printing process started by importing the Autodesk Fusion Model to the PreForm software, where parameters were defined. Once the printing ends, support structures are removed to restore precise geometry. After this process, the part is cleaned in isopropyl alcohol to remove excess resin and then subjected to a controlled post-cure under UV light at 60°C for 30 minutes, which drives the polymerization to completion and locks in the final material properties [27].

Complementing the black resin V4, Formlabs Tough 2000 resin was selected for the integrated screw–spring subassembly because it delivers the precise combination of toughness, elasticity and dimensional stability required for its printing. With an elongation at break of 48 % [28], tensile modulus of 2.2 GPa and flexural modulus of 1.9 GPa [28], Tough 2000 resin was initially considered the ideal material for a fully printed screw–spring assembly. Its apparent high ductility was expected to allow the spring to compress repeatedly without cracking, and its stiffness and toughness were expected to keep the threads dimensionally stable.

In addition, the construction of the integrated screw-spring assembly began in the PreForm software also, where the material profile was set to Tough 2000 V1. The part was tilted approximately 30° on the build platform to minimize support interference on the delicate spring coils while preserving thread accuracy. During printing, the resin vat was maintained at around 33 °C to ensure consistent viscosity and once printed a UV laser cured each successive layer and then the components were immersed in isopropanol. Finally, the cleaned parts underwent to the support removal process. This inverted sequence as compared with the bottom-casing part, was adopted due to the structural fragility of the ultra-thin screw. Performing a post-curing step prior to immersion in IPA enhances the mechanical stability of the polymer matrix, ensuring that the spring elements retain their geometry during the cleaning process.

2.4.4.2 Thread actuated pressure mechanism

Building on the spring-loaded prototype, the refined design maintains the pressure condition recommended by ICFO, where the adjustment screw compresses the spring and the infant’s scalp pushes back against it, automatically modulating the force on the optode.

This new version of the mechanism comprises four main elements: an adjustment screw, a top housing, a spring and the bottom housing. The adjustment screw features a threaded shaft that threads into the top housing. The top housing is a cylinder with an internal thread that guides the screw’s travel. The bottom part holds the spring and both include a central hole to accommodate the optode tip. When assembling the upper and the bottom parts, there is a side opening for the optode’s fibre.

During functionality, the researcher has to rotate the adjustment screw, causing it to advance downward and automatically the spring in the optode tip is compressed. Once the assembly is seated on the infant’s head, the optode tip makes contact with the scalp; at that moment, the compressed spring reacts by pushing back upward. Because the spring’s force and the scalp’s resistance are balanced, the result is a self-regulating, constant pressure. This ensures the optode tip maintains reliable contact with the scalp without exerting excessive force or causing discomfort.

This refined mechanism preserves the rapid setup workflow of the previous spring-loaded mechanism. Upon the family’s arrival, one researcher assesses the infant’s hair density and uses that information to pre-adjust each optode holder, while another explains the experimental protocol to the parents. By the time the infant enters the testing room, the probes are positioned on the scalp and secured with the EEG cap. The optodes are ready for immediate signal assessment, and any fine-tuning of contact pressure can be performed in seconds.

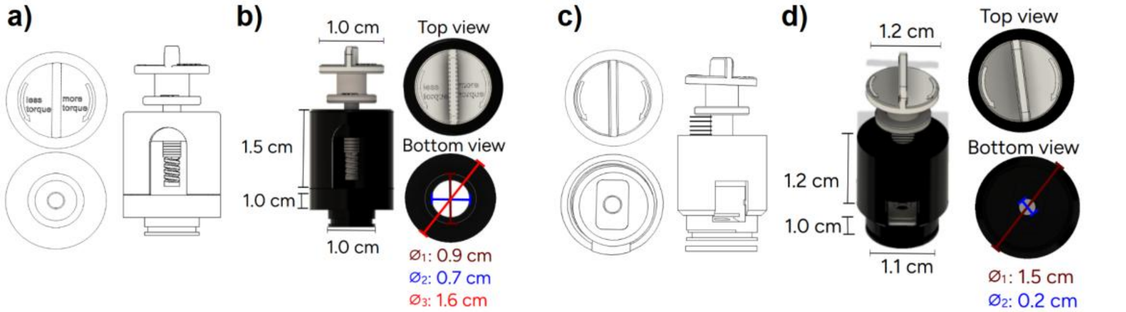

Figure 6. Comparison between the two printed mechanisms. (a) 2D sketch of the spring-loaded design. (b) 3D model of the spring-loaded mechanism in Fusion 360. (c) 2D sketch of the threaded actuated design. (d) 3D model of the thread actuated mechanism in Fusion 360. The spring in (a, b) was replaced by a flat spring positioned at the bottom in (c, d). The screw head engravings in (a, b) were removed in (c, d) due to print resolution limits. The bottom hole was redesigned in (c, d) to match the exact tip size of the optode, and overall dimensions were adjusted to improve stability, precision, and fit.

After multiple part failures using the Formlabs 3 resin printer, fabrication was shifted to Computer Numerical Control (CNC) turning for improved strength and precision. The screw, the casing and bottom part were machined on a Haas ST-10 CNC turning machine, starting with a 3D model designed in Autodesk Fusion 360. This model was then exported as a STEP file, imported in CAD, aligned along the Z-axis (lathe rotation axis) and prepared using CAM tools.

The parts were machined from a bar from a polyamide 6.6 (nylon) reinforced with 30% glass fibre (PA6.6 + GF30 bar). A machining JOB was configured by selecting the part, defining the bar stock dimensions, setting safety margins and assigning the coordinate origin (G54). Subsequently, 4 sequential operations were applied: roughing to remove bulk material, contour finishing for final dimensions, threading for the screw, and drilling for the central hole. The entire toolpath was simulated to ensure safe and accurate execution, then post-processed to generate G-code, a standardized programming language used to control CNC machines, specific to the Haas ST-10, the CNC machine used for this operation. After machining, the components were washed in isopropanol and polished to achieve the final surface finish.

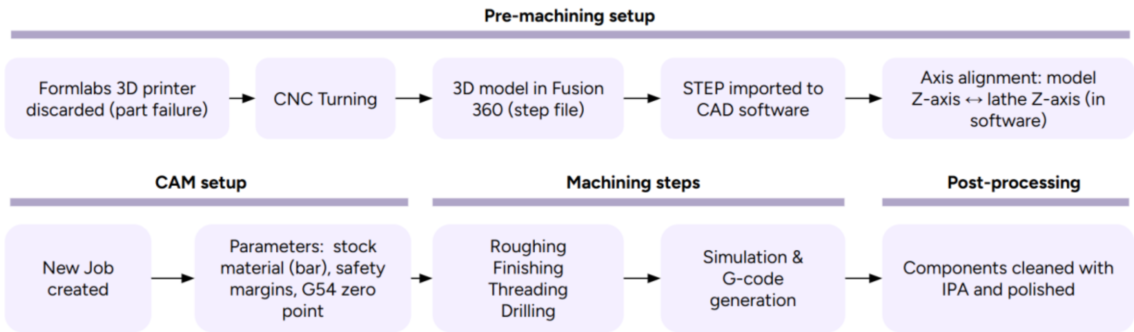

Figure 7. Workflow of CNC fabrication for mechanical parts, from model design to post-processing.

The spring geometry, however, was 3D printed using the Bambu Lab X1E printer. The chosen material was a carbon fibre reinforced polylactic acid (PLA-CF). This filament was selected because its added carbon fibres minimise warping and give a crisp matte surface, providing the dimensional stability and surface finish required for a low-load spring.

The PLA-CF filament was loaded into the Bambu Lab X1E printer, which performed automatic bed levelling and flow calibration. The part was then printed layer by layer at 220 °C. The printed part was then removed, cleaned with isopropanol and air-dried before use.

2.5 Development of the multimodal probes

This phase focused on the design and development of a spatial distribution integrating sensors, along with their associated wires and fibres, from three different modalities. The resulting configuration, referred to as the multimodal probe, organises EEG electrodes as well as fNIRS and fDCS optodes including the fDCS pressure mechanism over the ROI.

Designing a safe and functional system required consideration of infant anatomical dimensions. The cephalic perimeter, that is the head circumference, is a parameter that defines the surface available for sensor placement. It is measured by placing a measuring tape around the widest part of the head, typically from the forehead just above the eyebrows to the most prominent part at the back of the skull. According to World Health Organization, the average for 4- to 5-month-old infants ranges from 40.5 to 42.8 cm in boys and 39.3 to 41.8 cm in girls [29].

To target the temporal region involved in early language processing [30], rectangular lateral probes (7.0–7.8 cm wide, 4.8 cm high) were designed, along with a 4.0 × 1.0 cm lateral support for attachment to the ear, ensuring stability and proper placement. These dimensions balance anatomical constraints and technical needs for fDCS/fNIRS integration, as detailed below. Additionally, an occipital probe was included as a control region, with designs ranging from 5.0 to 7.0 cm in width and 2.5 to 5.0 cm in height.

Another important constraint was the SDS, the linear distance between a light source and its detector, directly affects photon penetration depth and therefore, system sensitivity to different tissue layers [31]. Greater SDS increases sensitivity to cortical signals but leads to higher attenuation and signal dispersion [32].

In multimodal systems, ensuring that they measure the same area requires aligning their respective SDS midpoints, which represent the area of the brain being measured, critical for comparing data across modalities. Therefore, the SDS must be carefully chosen to balance signal strength and spatial specificity by targeting the cortical region of interest.

2.5.1 Systematic workflow for probe design

The probe development pipeline began by generating 2D distributions. Designs that demonstrated optimal characteristics were promoted to the 3D modelling stage, and those that passed this evaluation were ultimately printed.

To address the spatial constraints, a custom Python script was developed to simulate and evaluate multiple spatial configurations in a two-dimensional layout representing the cranial surface of a 4- to 5-month-old infant.

Each sensor type was defined with its actual physical diameter (EEG: 1.1 cm, fNIRS detector: 1.1 cm, fNIRS source: 0.7 cm, fDCS: 0.2 cm). The script implemented minimum and maximum SDS thresholds to validate the positioning of optical channels: 1.8-3.0 cm for fNIRS and 1.5-2.0 cm for fDCS. These ranges were selected based on reported values [13], [33] and [12] and prior internal protocols, serving as a point for design exploration.

The script computed all pairwise distances between sources and detectors, identifying valid optical pairs based on SDS constraints. For each, it computed the midpoint, representing the scalp measurement area. Although fNIRS and fDCS pairs may differ slightly in surface location, their midpoints reflect the same evaluated cortical regions due to the banana-shaped sensitivity profile of both techniques. Therefore, as long as SDS values are within the defined range, the signals are spatially comparable and originate from functionally equivalent brain tissue. This tool has proven functional for both the lateral probes and the occipital one.

The code generated layout visualizations with color-coded sensor markers and automated annotations for valid pairs, outputting each pair's SDS and midpoint. This enabled rapid assessment of spatial alignment and signal quality. Accordingly, by automating geometric validation, the tool streamlined design iterations, eliminating the need for manual 3D evaluation. This early-stage optimization ensured that only viable configurations progressed to 3D modelling, contributing to the development of a high-performance system. The most relevant distributions guiding final designs are shown in *Figure SI- 17*.

Once the final spatial distribution was defined, the layout was transferred into 3D models developed in Autodesk Fusion 360. An essential challenge in this phase involved routing the fragile fDCS optical fibres, which are highly sensitive to bending, tight curvature and compression, conditions that can compromise signal transmission. Therefore, the 3D design required careful path planning to prevent overlap and ensure each fibre followed an independent trajectory with minimal mechanical stress. As a result, some layouts that were feasible in 2D, proved unsuitable in 3D due to fibre routing constraints, prompting iterative redesign and reassessment. Although many configurations were tested computationally, only the most viable were modelled in 3D.

2.5.2 Multimodal probes retained in the design phase

Among the configurations developed during the design phase in the Python script, 5 lateral probe designs were done: 2 remained in the design phase, while 3 advanced to 3D modelling and were successfully printed. Additionally, 1 occipital probe was both designed and printed.

The first validated 2D-layout, occupying a 4.8 × 7.8 cm area, included 4 EEG electrodes, 3 fNIRS detectors, 3 fNIRS sources, 6 fDCS detectors and 2 fDCS sources (see *Table SI-1* for sensor placement and *Figure SI- 18* for sensor distribution). Although 6 regions were measured by fNIRS and fDCS simultaneously, one region was exclusively monitored by fNIRS and two others only by fDCS. Despite this, the design was modelled in 3D to assess mechanical feasibility, particularly for fibre routing, not evaluable in 2D.

The second configuration, reduced to 4.8 × 7.0 cm for improved fit, included 4 EEG electrodes, 2 fNIRS detectors, 2 fNIRS sources, 1 fDCS source, and 4 fDCS detectors. It allowed for 4 fNIRS–fDCS measurement areas while maintaining appropriate spacing between optodes, overcoming critical limitations of the first design.

2.5.3 Design and construction of the printed probes

This section presents the development of the multimodal probe configurations that were successfully 3D printed and adapted to incorporate the optode pressure mechanism. Printing was carried out on a Formlabs 3 using Flexible 80A resin. The 3D models were first prepared in the PreForm software, where printing parameters were defined, and then transferred to the printer. Once printed, the probes were washed in isopropanol to remove uncured resin, dried, and exposed to UV light to complete polymerization. After curing, support structures were carefully removed, and a uniform black coating was applied to enhance light absorption and ensure optimal optical performance.

The first printed probe corresponds to the third configuration approved in 2D, a refined version of the previous layout, incorporating a 0.9 cm diameter opening to accommodate a mechanical locking system that ensures secure attachment of the spring-loaded component. It preserved all critical constraints, including SDS values (2.20 cm for fNIRS, 1.80 cm for fDCS), proper routing of wires and fibres, and multimodal overlap across four measurement areas. The design was validated in 2D, modelled in 3D and printed.

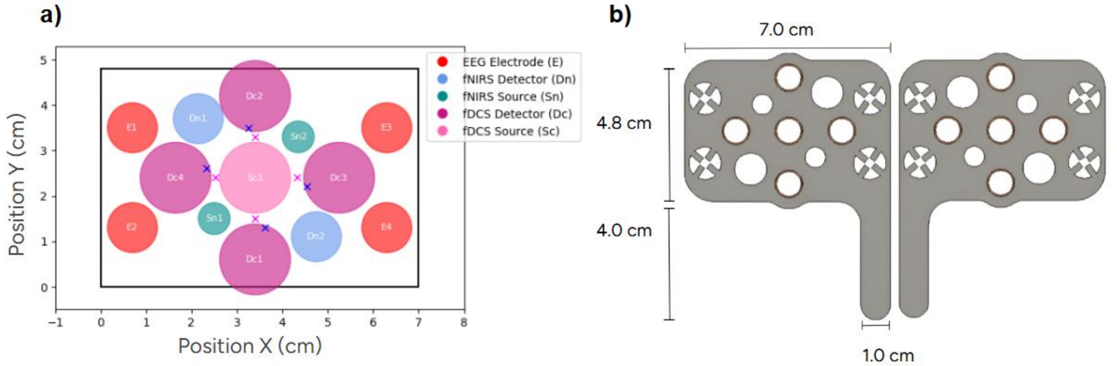

Figure 8. a) Schematic layout of the multimodal sensor generated through the Python code. b) 3D model of the first printed probe (left and right hemisphere) with hole distribution in Autodesk Fusion 360.

Building on this prototype, the design advanced to another configuration to accommodate the final version of the optode pressure mechanism, increasing the diameter of fDCS holes from 0.9 cm to 1.5 cm. Sensor positions were optimized to achieve SDS values of 2.55–2.7 cm for fNIRS and 1.8 cm for fDCS. After validating spatial constraints in 2D using the Python script, the configuration was modelled in Autodesk Fusion 360.

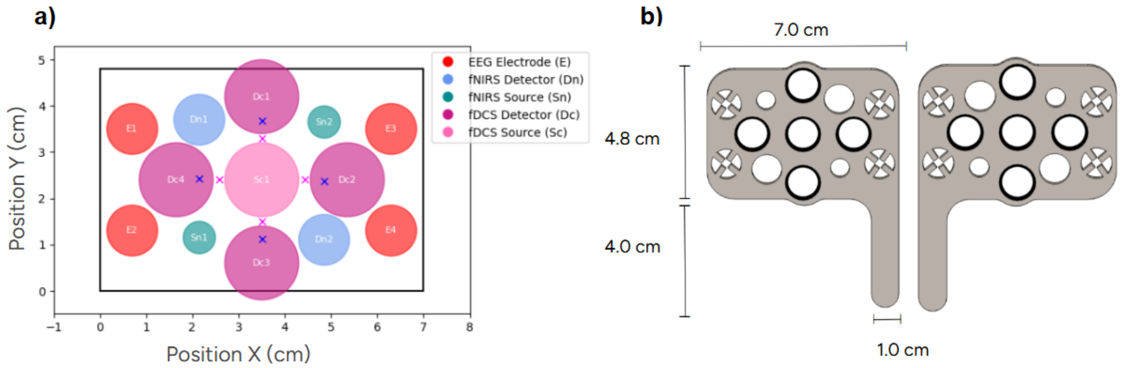

Figure 9. a) Schematic layout of the multimodal sensor generated through the Python code. b) 3D model of the second printed probe (left and right hemisphere) with hole distribution in Autodesk Fusion 360.

From this point, one last redesign yielded the final lateral multimodal probe. It builds on the previous one, addressing prior limitations by expanding the area to 4.8 × 7.8 cm, while still fitting the infant's head. This adjustment solved fibre bending issues and ensured accommodation of all components. It integrates space for 4 EEG electrodes, 2 fNIRS sources, 2 fNIRS detectors, one fDCS source and 4 fDCS detectors. It maintains 4 measurement areas, with both fNIRS and fDCS optodes targeting the same regions, enabling multimodal acquisition.

For fDCS channels, an SDS of 1.80 cm was selected, based on a prior EEG–fDCS study conducted by the SAP group [Ghailan et al., in preparation], where a 1.5 cm SDS yielded reliable signals. The slight increase in the new design accommodates the spatial requirements of EEG, fNIRS, and fDCS without compromising signal quality. To define the optimal SDS for fNIRS, studies involving infants were reviewed [13], [12], [33], with age being an essential factor due to developmental differences in head size and hair.

Studies in 3–5-month-olds using a 2.5 cm SDS [13] reported high data retention (2/31 excluded), while [12] used 2.0–3.0 cm in 4-month-olds with 7/49 exclusions. In contrast, [33] used 3.0 cm in newborns and excluded 6/24. Given that the present system targets 4–5-month-old infants, more weight was given to [13] and [12]. Considering cortical sensitivity, data quality and spatial integration of all modalities, an SDS of 2.90 cm was selected for fNIRS. Following 2D validation in Python, the probe was modelled in 3D using Autodesk Fusion 360 and printed to confirm mechanical fit and functionality.

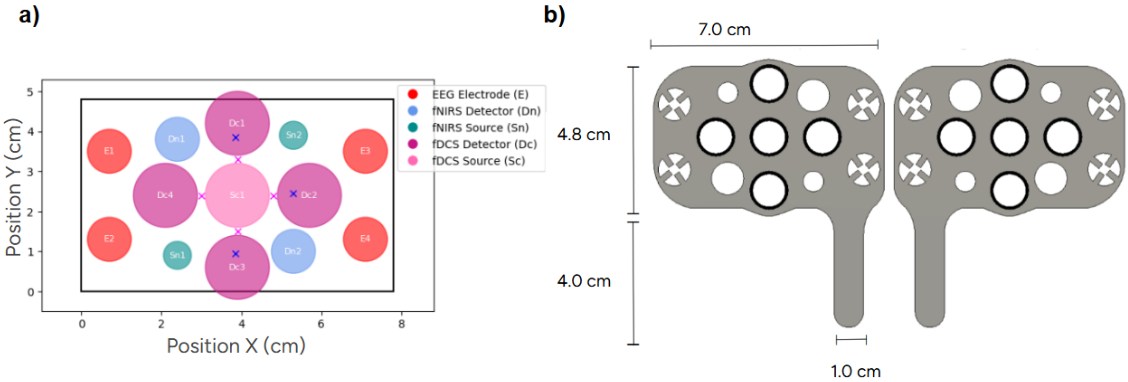

Figure 10. a) Schematic layout of the multimodal sensor generated through the Python code. b) 3D model of the final lateral probe (left and right hemisphere) with hole distribution in Autodesk Fusion 360.

To complement the final setup, an occipital probe was designed to serve as a control region within the multimodal neuroimaging system. Among several initial proposals, one configuration was modelled due to its compatibility with the overall setup. It measures 7.0 cm in width and 5.0 cm in height, and includes 1 fNIRS source, 1 fNIRS detector, 1 fDCS source, 1 fDCS detector, 2 EEG electrodes and a rectangular accelerometer.

The accelerometer was incorporated to monitor head movements and improve the reliability of the acquired signals [34]. Since the occipital region is not expected to show activation during language-related tasks [17], it provides an ideal location to detect motion artifacts without interfering with functional data from target areas. By capturing real-time movement information, the accelerometer allows researchers to distinguish between true neural signals and noise caused by infant motion.

To guarantee consistency across all probes, the SDS were kept identical to those in the lateral configuration: 1.80 cm for fDCS and 2.90 cm for fNIRS. This choice enables direct comparison between active and control regions under the same acquisition conditions, helping confirm that any observed differences in signal arise from neural activation rather than design variability.

Given its role as a control, the probe features a low density of optodes, enough to validate the system and noise characteristics without overloading the design. The sensor layout was first optimized in 2D using Python and subsequently modelled in 3D using Autodesk Fusion 360 to ensure precise fit, wire and fibre routing, and component integration.

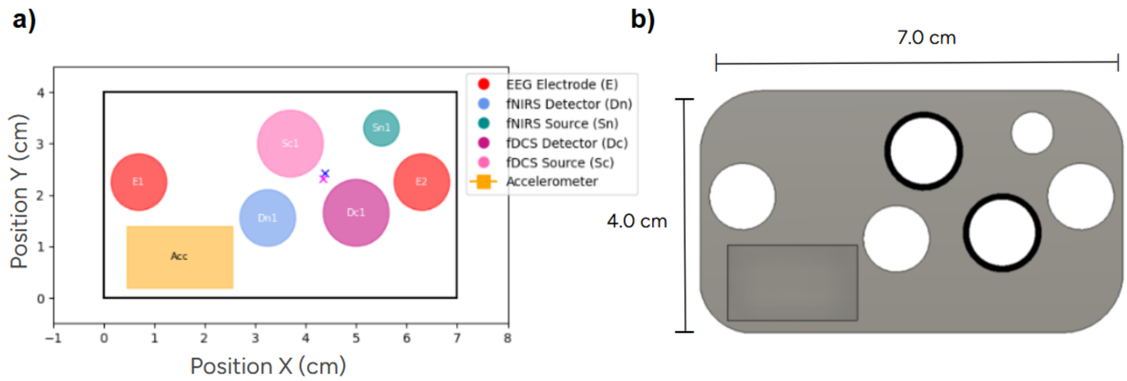

Figure 11. a) Schematic layout of the multimodal sensor generated through the Python code. b) 3D model of the occipital probe with hole distribution designed in Autodesk Fusion 360.

2.6 Validation of the pressure mechanism using a functional probe layout

A validation of the optode–pressure mechanism with one probe was conducted on a young adult volunteer with dark brown dense hair and had a head circumference of 59.4 cm. Although the focus of this project is on infant neuroimaging, the validation test was conducted on an adult to avoid the ethical and logistical constraints associated with testing on infants. Performing such a trial with an infant would have required prior approval from an ethics committee, as well as written informed consent from the parents, procedures that were not feasible within the project timeline. In contrast, conducting the test on an adult only required verbal consent, making it a practical and compliant solution for preliminary technical validation.

The occipital probe was positioned over the right temporal region. Although it is designed for the occipital cortex, this probe was selected because it provided one opening for an fDCS source and one for a detector, the minimal configuration required to evaluate the effect of the pressure mechanism on light intensity with a SDS of 1.8 cm. Using this compact setup avoided the need for a larger, denser probe with multiple optodes, which would have been unnecessary for the purposes of this test.

The main objective of the evaluation was to verify whether the pressure mechanism improved the photon counts per second of the measured signal. To this end, one source and four fDCS channels integrated in a single shared detector fibre were recorded. The temporal region was chosen as it corresponds to the intended target area of the final multimodal system.

The probe was secured using a bandage that covered the entire structure, leaving the area around the pressure screws accessible for adjusting. As no adult-size silicone mesh EEG cap was available in the laboratory, it was not possible to test the mechanism within the full setup. Such a mesh cap would have been suitable, as it is the one selected for the infant study and its flexible design and perforations allow local adjustment and access to the screw. In contrast, the rigid structure of standard EEG caps would have interfered with the vertical adjustment of the pressure screw, making them incompatible with the mechanism.

3. RESULTS

This section presents the outcomes derived from each stage of system development, including the validation of EEG data acquisition to assess its compatibility with simultaneous EEG-fDCS recordings. Each step built upon the findings of the previous one, forming a structured progression towards a setup.

3.1 EEG data analysis from the visual study

To assess the quality of EEG recordings during concurrent fDCS stimulation, the frequency spectra of signals recorded over occipital electrodes were analysed while participants were exposed to a 5 Hz checkerboard pattern.

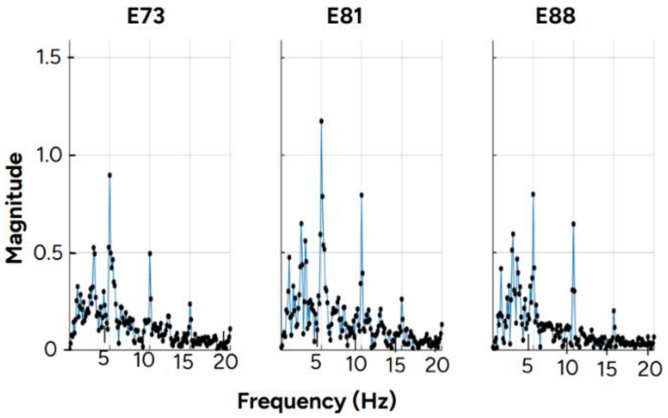

Figure 12. Fast-Fourier Transform spectra at electrodes E73, E81 and E88 of a 5 Hz stimulus for a signal duration of 8 seconds. A peak can be observed at 5 Hz, along with its harmonics at 10 Hz and 15 Hz.

The results in *Figure SI- 2* show how the frequency spectrum of a 5 Hz signal changes depending on the duration of the recording. When the signal is only 1 second long, as shown in *Figure SI- 2*, the peak at 5 Hz is wide and not well defined, meaning that it is harder to distinguish real components from noise. As the duration increases, the peak becomes sharper and narrower.

In addition to the main peak at 5 Hz, there can be observed consistent peaks at 10 Hz and 15 Hz across all signal durations. These are harmonics of the frequency stimulation. A further peak at approximately 3 Hz is also present; although its origin remains unclear, it may correspond to some endogenous low-frequency oscillations commonly reported in infants of this age [35]. Moreover, the frequency spectra from electrodes E73, E81, and E88, all located over the occipital region associated with visual processing, consistently showed 1 peak at the stimulation frequency (5 Hz) and in the first and second harmonics.

3.2 Preliminary conceptual analysis of the pressure mechanisms

Four preliminary designs were discarded for different reasons. First, the screw gear mechanism revealed that demanded a voluminous enclosure to house both the gear and zipper. This exceeded the spatial constraints of the probe, reduced available surface for sensors and added excessive weight, rendering it unsuitable for infant use. In parallel, spatial analysis revealed that the lateral expansion required by the dual screw mechanism, due to the horizontal screw, exceeded the acceptable dimensions of the probe. Moreover, the magnetic mechanism was dismissed based on theoretical analysis. Although potentially effective for applying pressure, the required magnetic strength was predicted to generate EEG artifacts [36], introducing a significant risk to signal quality. Finally, the vertical piston design was also rejected due to practical limitations. The need for multiple internal levels complicated assembly, and the lack of gripping features made precise positioning during setup difficult. These findings highlighted the need for a more compact, lightweight and functional solution.

3.3 Early evaluation of the pressure mechanism prototypes

Prior to *in-vivo* measurements, the three pressure mechanism prototypes were tested on an infant mannequin wearing an elastic neuroimaging cap, as the final probe was not yet available. These tests focused on the qualitative assessment of mechanical stability, pressure distribution and adaptability to curved surfaces.

The lever-based mechanism (see *Figure SI- 10*) showed mechanical stability but presented limitations. First, its multiple components increased assembly complexity. Moreover, the restriction to three fixed pressure levels reduced adaptability to different hair conditions. Finally, its bulky design raised concerns about infant comfort and parental acceptance. While discarded, it provided guidance on structural support and usability, which influenced later iterations. On the other hand, the modular compression system (see *Figure SI- 12*) was also discarded after qualitative testing revealed that achieving lower compression levels required excessive manual force and imposed continuous pressure on the head. The lack of dynamic adjustability introduced a risk of discomfort and limited its feasibility for infant application.

In contrast, the spring-loaded mechanism (see *Figure SI- 13*) successfully demonstrated controlled vertical pressure application, validating the functional concept. However, it failed to apply counter pressure from the scalp. Despite this limitation, the prototype established a solid foundation for the final mechanism, from which key design elements were retained.

3.4 Evaluation of the printed pressure mechanisms

The two pressure mechanisms fabricated were evaluated to assess their mechanical performance and potential impact on signal quality.

During the 3D printing phase, the spring-loaded model (see *Figure 14 – a,b*) faced critical issues that prevented further validation. Although Formlabs Tough 2000 resin was chosen for its high ductility and stiffness, the printer's resolution was insufficient to reproduce the fine geometry required for the screw–spring system. Moreover, the printed springs were rigid, lacked elasticity and proved extremely fragile, breaking with minimal handling. These mechanical failures were combined with poor printing of the text written in the top, as its tiny size exceeded the resolution of the printer. Since the spring and thread were intended to be printed as a single unit, these issues rendered the prototype non-functional and confined the evaluation to the printing stage.

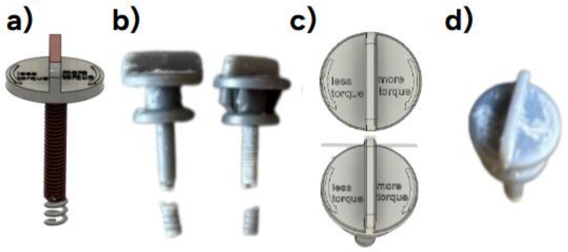

Figure 13. Comparison between the intended 3D designs and the printed prototypes. (a) Autodesk Fusion model of the integrated screw–spring mechanism. (b) printed mechanism, whose springs were fractured. (c) Autodesk Fusion model of the top of the screw showing the "less torque / more torque" legend. (d) printed component, in which the text failed because its feature size fell below the printer's resolution.

The thread actuated pressure mechanism performed reliably during assembly. The process to assembly the mechanism is by introducing the tip inside the spring, which is then inserted into the bottom housing. The outer casing closes over this assembly, and the adjustment screw is turned to apply pressure on the optode. The bottom housing aligns perfectly within the probe body, allowing the optode tip to pass through the mechanism as designed. Its black colour is prepared to absorb any stray light, minimizing unwanted reflections and preserving signal integrity. These observations are exclusively of the mechanical assembly and fit of the pressure mechanism. The assessment of its impact on signal quality, using the full setup, is presented section *3.7 Validation of the pressure mechanism using a functional probe layout.*

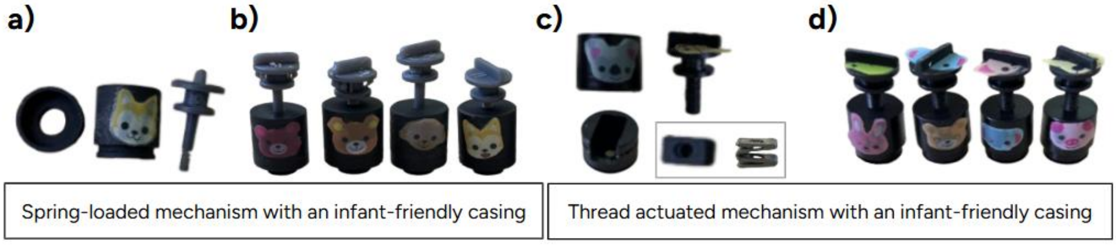

Figure 14. 3D printed mechanisms. (a) Components of the spring-loaded mechanism: bottom part, casing and the failed screw-spring unit. (b) Same spring-loaded mechanism assembled in its intended configuration. (c) Components of the thread actuated mechanism: a bottom part, lateral and frontal view of the new flat spring (note the change from a coil to a flat shape compared with a), a smaller casing and a redesigned screw. (d) Threaded actuated mechanism fully assembled.

3.5 Evaluation of the unprinted probe designs

Two lateral probe configurations reached the modelling stage but were not printed due to critical limitations. The first layout was discarded for two main reasons. Its optode distribution provided incomplete multimodal overlap, with regions covered exclusively by fNIRS or fDCS, adding unnecessary complexity, weight and spatial density without contributing to shared measurement areas, an essential requirement for infant applications. Additionally, 3D modelling revealed sharp bending and overlapping of optical fibres, posing a high risk of breakage and signal loss.

The second layout met spatial and functional criteria but was not printed due to the absence of the pressure mechanism. The 3D model included holder apertures but without fixation features, and fibre integration was postponed.

3.6 Evaluation of the printed probe designs

All probe designs that advanced to the 3D modelling stage were subjected to proof-of-concept trials on an infant mannequin head measuring 41 cm in diameter, the average head size of a 4–5-month-old. In each test, the fully assembled probes (including optode pressure mechanisms, optical fibres, and wiring) were evaluated for conformal fit, mechanical stability and routing integrity.

The first printed probe with Flexible resin 80A was assembled with the optode pressure mechanism and all optical fibres and wiring. First, all the components fit within the probe, confirming the precision of the 3D design. When tested on the infant mannequin the probe conformed perfectly to the head form. The flexible resin accommodated the mannequin's curvature, and the dimensions proved ideal for infant application. This fit validated the choice of Flexible 80A resin and confirmed that the material would be retained in the final design. However, the design was discarded when the optode pressure mechanism increased its bottom diameter from 0.9 to 1.5 cm, requiring a redesigned integration.

Based on the updated mechanism, a second probe was generated using the Python script and developed in Fusion 360, where the design revealed that routing the fDCS fibres would require bending due to limited structure space. This presented a concern, as optical fibres can easily be damaged if bent. Nevertheless, it was printed to assess the component integration, confirming that the new mechanism could be integrated into the structure.

During proof-of-concept trials on an infant mannequin with a 41 cm head circumference, the assembled probe, including the pressure mechanism adapted well to the head shape and all structural elements fit as intended. However, as anticipated from the 3D model analysis, the fDCS fibres experienced bending, risking fracture. Consequently, this configuration was discarded, and its insights guided the final probe geometry refinement.

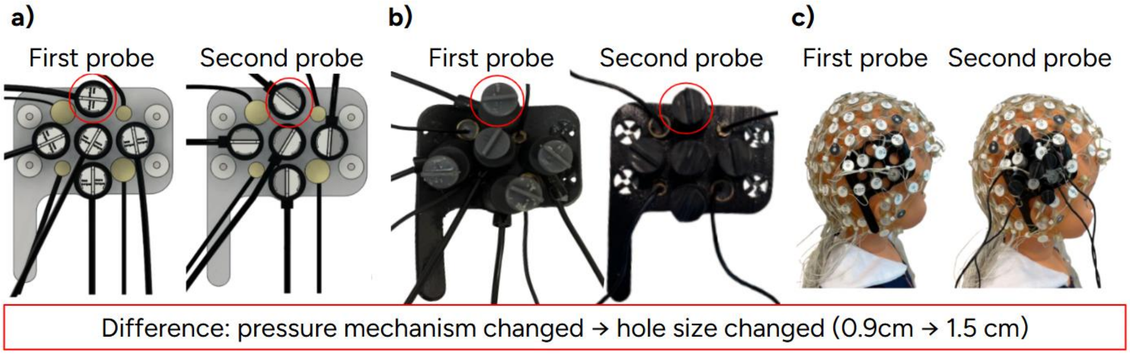

Figure 15. Comparison of the first and second lateral probe designs. a) 3D models in Fusion 360. b) Fully assembled prototypes. c) Probes mounted on an infant mannequin. The two versions differ only in the diameter of the opening for the pressure mechanism (5 different holes). The second design has inadequate fibre routing that forces fDCS fibres to bend, prompting the development of a third, redesigned probe.

In the final probe iteration, to accommodate the new pressure component and to have enough space for all the fibres and wires, the lateral probe width increased by 8 mm, with respect to the previous probes (4 mm on each side). However, coverage of the temporal region remains unchanged because the support leg is positioned identically to that of the fourth lateral probe model. Moreover, all fDCS fibres follow straight trajectories within the probe, without bending, eliminating fracture risk. Additionally, multimodal overlap is verified across all four target regions: each region receives both fNIRS and fDCS measurements exactly as intended, confirming that design adjustments maximize measurement zones without compromising anatomical specificity. Finally, testing on the infant mannequin, demonstrated the elastic adaptation of the probe material on an infant's head, ensuring a proper fit.

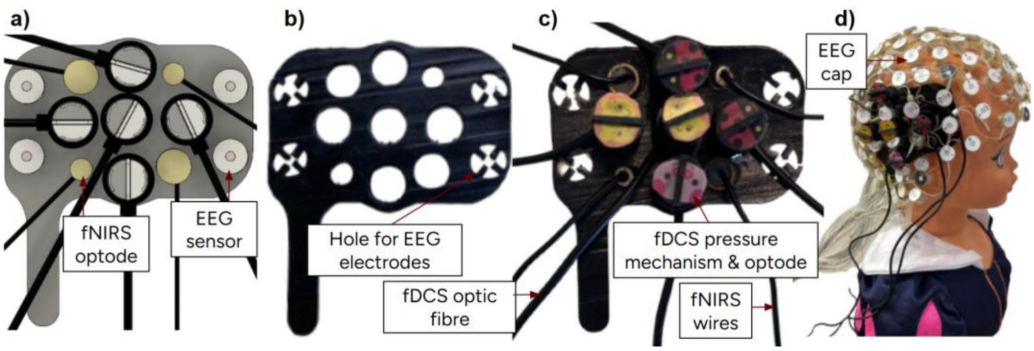

Figure 16. Final lateral probe. a) 3D design illustrating successful fibre routing layout. B) 3D printed probe with dedicated slots for each element. C) 3D printed probe assembled with the thread actuated mechanism and fNIRS optodes and fDCS fibres. d) Infant mannequin including the EEG and the probe assembly.

The occipital probe, designed to follow scalp curvature, was validated on an infant mannequin. Fibre routing remained straight to avoid stress points. In addition, SDS match the lateral configuration (1.80 cm for fDCS, 2.90 cm for fNIRS), establishing the probe as a robust control region under identical acquisition parameters. It also accommodates the accelerometer within its slot, with the protective cover fully enclosing the electronic board. This arrangement aims to protect the accelerometer during EEG preparation.

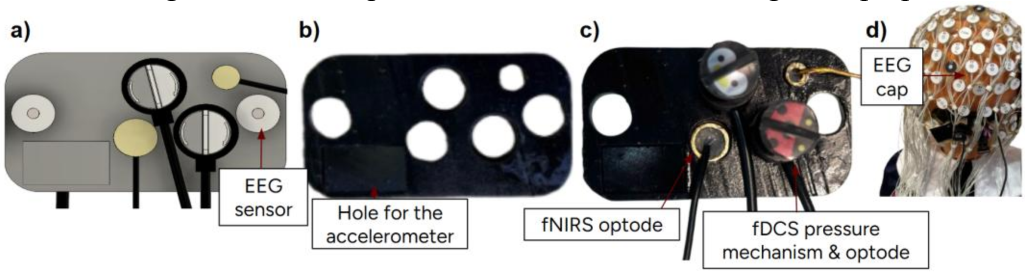

Figure 17. Occipital probe. a) 3D design illustrating successful fibre routing layout. b) 3D printed probe with dedicated slots for each element. C) 3D printed probe assembled with the thread actuated mechanism and fNIRS optodes and fDCS fibres. d) Infant mannequin including the EEG and the probe assembly.

3.7 Validation of the pressure mechanism using a functional probe layout

A test was carried out to quantify the influence of the pressure mechanism on fDCS signal intensity. A laser source of 785nm, at power of 19.3 mW, and four detectors configured as independent channels were mounted on the right temporal region, as explained in section *2.6 Validation of the pressure mechanism using a functional probe layout.*

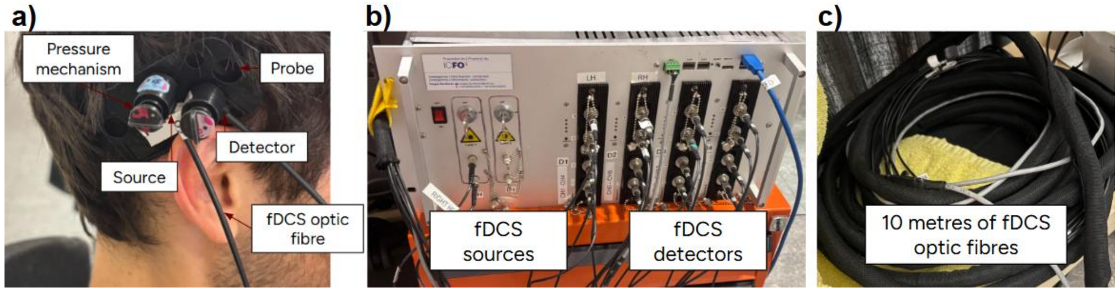

Figure 18. Experimental setup for the validation of the fDCS pressure mechanism at the Neuroscience Laboratory of UPF. a) Probe mounted on the right temporal region. The system includes a source and a detector, secured with an individual pressure mechanism to enhance scalp contact. b) fDCS acquisition system of ICFO, showing the sources and the detectors modules used to record intensity across channels. c) 10 metres of optic fibres used to connect the source and detector optodes to the acquisition device.

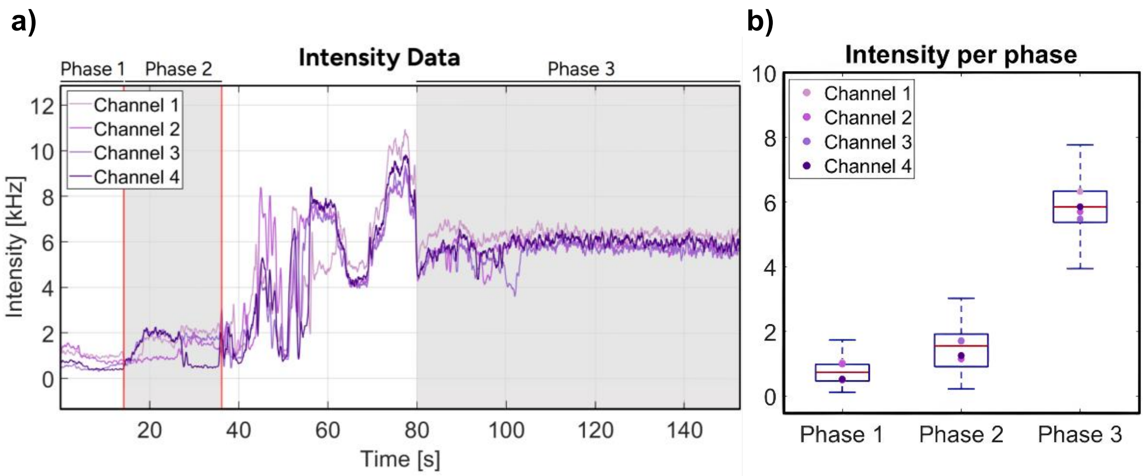

Figure 19. Intensity recorded across fDCS channels using the pressure mechanism. a) Time series, showing four phases. Phase 1: no pressure, Phase 2: compression of the source, Transitional phase: full compression of both source and detector, and: Phase 3 stabilization. b) Box plots summarising signal intensity per phase.

The intensity traces for all four channels, shown in *Figure 19 – a*, reveal four distinct phases. In the initial period (Phase 1, 0–14.15 s), no pressure was applied and the optodes rested passively on the hair, producing signals of lower than 2 kHz, around 1KHz as expected from light room contamination. During Phase 2 (14.15–36.09 s), the screw beneath the source optode was incrementally lowered, yielding a modest rise in intensity. The subsequent interval (36.09–80 s) corresponds to the progressive tightening of the detector screw and the signal amplitude increased continuously while contact improved, but this transient segment was excluded from further analysis as it was a transient phase. Once both screws were fully engaged, the system entered a stable regime (Phase 3, 80–150 s) in which intensity settled at approximately 6.5 kHz across all channels.

Moreover, *Figure 19 - b* displays box-and-whisker plots of signal intensity for the four fDCS channels across Phases 1, 2, and 3. In Phase 1, all channels show consistently low intensity values, clustered around 1 kHz. In Phase 2, a slight increase is observed across channels, though the change is modest and variability remains minimal. In Phase 3, all four channels show a marked increase in signal intensity, with median values approaching 6.5 kHz. Small differences in magnitude between channels appear in this phase, reflecting channel-specific variation in optode–scalp contact despite the same overall setup. The data show a tendency of light increase over phases, suggesting progressive improvement in optode-scalp contact.

3.8 Final setup

In the lateral multimodal probes (2), four EEG electrodes (E46, E33, E26, E45) in the left hemisphere secure the setup to the infant scalp with no detectable displacement under gentle perturbation. These channels covered the whole temporal lobe, including the perisylvian region, the ROI for early language processing in this study. To mirror this area, four EEG electrodes (E102, E122, E2, E108) in the right hemisphere were selected. In the occipital probe, 2 EEG electrodes (E73, E95) secure the setup to the scalp as this region will be the control area of the study.

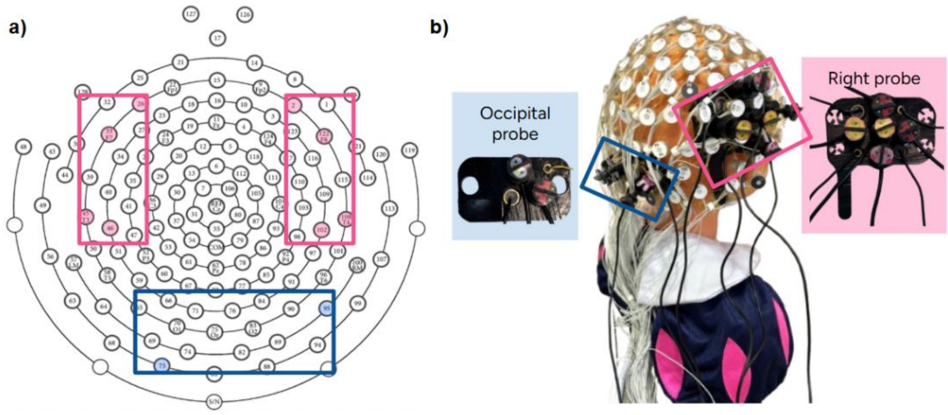

Figure 20. Multimodal setup. a) 128-channel EEG layout with the selected regions for probe placement: temporal (pink rectangles) and occipital region (blue rectangle). b) Infant mannequin wearing the EEG cap with the whole multimodal setup.

4. DISCUSSION

This Bachelor's Thesis set out to contribute to the development of a multimodal neuroimaging system for studying early language processing in 4-5-month-old infants. First, data quality and interpretability were enhanced through the reanalysis of previously acquired recordings. Second, a novel mechanical solution was developed to optimise scalp–optode contact for fDCS measurements, aiming to reduce signal loss associated with hair interference. Third, a complete sensor distribution was designed and prototyped, successfully integrating EEG, fNIRS, and fDCS components within a configuration that meets both anatomical and technical constraints specific to infant neuroimaging.

A first requirement was the selection of the optimal EEG and the verification that it would remain reliable within the combined framework. The analysis of EEG data from all 18 infants showed that using 8 second recording windows, aligned with the full duration of the 5 Hz checkerboard stimulus, produced the sharpest and narrowest peaks at the stimulation frequency. Longer windows enhance frequency resolution and aid separation of nearby spectral peaks. The results also indicate that a minimum of 4 seconds of visual stimulation are required to reliably detect the SSVEP response, with longer windows yielding progressively better detection due to the increased number of frequency bins. Additional peaks at 10 Hz and 15 Hz confirmed the presence of harmonics of the stimulus and demonstrated that the EEG system performed properly within the multimodal setup, reliably capturing brain signals even during concurrent fDCS measurements using the headband configuration. Finally, the almost identical patterns observed at electrodes E73, E81 and E88 were expected given that the occipital cortex tends to respond symmetrically for this type of visual stimulation [37]. This similarity across electrodes supports the interpretation that the stimulus evoked a reliable and coherent response in visual areas of the brain.

This re-analysis also fulfilled the first objective of the thesis: improving data usability through reanalysis of existing recordings. Due to the consistent differences in screen brightness across the baseline, attention-getter and checkerboard pattern, it was possible to reconstruct the stimulus timing even in cases where the mirror failed to provide confirmation due to suboptimal placement. This allowed recovering 23 participants who had initially been excluded. Although only 18 recordings ultimately met the final quality standards, this approach significantly increased the usable dataset and demonstrated the value of refined analysis for enhancing signal reliability in infant studies.

After the familiarization period, the focus turned to the other elements required to complete the system. The most decisive factor in reaching a multimodal setup with a pressure mechanism, was the iterative workflow that linked every stage of design, fabrication and testing. Starting with sketches, the project cycled through different layouts produced by the Python and then only the approved ones were developed in Fusion 360, followed by successive 3D printing. Each loop exposed a different bottleneck; excessive fibre bending, insufficient pressure range, materials printed with low resolution, broken components, and each failure informed the next refinement in geometry, material or mechanical tolerance.

This iterative refinement process was especially critical in the development of the pressure mechanism. The pressure mechanism underwent successive design iterations, beginning with conceptual designs, and proceeding through early prototypes tested on an infant mannequin, to optimize assembly simplicity, pressure adjustability and stability.

Compared with other fNIRS holders, such as NIRx [21], which require selection among pressure levels, the continuous adjustability in the new mechanism allows hair type and head shape variation to be addressed without swapping parts. This not only streamlines the workflow but also reduces the total number of spare components needed in the laboratory, minimizing the risk of losing components. Moreover, the threaded adjustment provides gradual, finely controlled pressure increases, ensuring smooth and gentle application. Furthermore, the black colour prevents stray reflections and the addition of small animal stickers on the casing soften the appearance, helping to reassure and comfort parents throughout the study.

Together, these advantages, rapid setup, minimal component handling, precise and passive pressure control with the spring and infant-friendly aesthetics, represent a significant improvement over existing commercial solution. Through this study, it has been demonstrated how iterative prototyping led to a pressure mechanism that maximizes usability, safety and signal reliability within the multimodal framework.

In parallel with these mechanical advances, and drawing on the same iterative philosophy, computational tools were deployed to streamline the geometric constraints of the probes. Designing multiple probe layouts, whether drawing on a paper or sketching in the computer, requires investing lot of time designing lots of layouts, computing the SDS for each pair of optodes in each configuration and computing each midpoint referring to the different measurement area. The python script generated every candidate layout in seconds while automatic computing SDS and midpoints. Early detection of spatial conflicts avoided multiple failed drawings and computations.

Once the 2D geometry was validated, 3D modelling of fibre trajectories revealed spatial conflicts before any physical printing. Detecting and correcting these issues saved material and time, as failed prints on the Formlabs can consume hours and material.

This emphasis on automated layout and digital prototyping underscores the essential role of computational tools. The rapid feedback loop of design, analysis and refinement not only reduced development time and costs but also delivered robust probe model.

As a result of this process, the final lateral and occipital probes achieved full multimodal overlap while eliminating fibre bending, thus meeting the one of the objectives of this BT, that was designing and developing a sensor distribution integrating EEG, fNIRS and fDCS optodes while respecting anatomical and technical constraints. Compared with earlier iterations, widening the lateral probe by 8 mm provided the clearance necessary for straight fibre routing without reducing temporal lobe coverage. Moreover, the identical SDS (2.90 cm for fNIRS, 1.80 cm for fDCS) across probes (lateral and occipital) ensure that any differences observed in subsequent infant recordings will reflect physiology rather than layout. Mechanical tests on the infant mannequin verified conformal fit, stability under gentle perturbation and unobstructed access to the pressure screws.

Complementing this evaluation, a practical test was conducted on an adult volunteer provided a practical approximation of hair-related signal attenuation. This test directly addresses the second objective of the project, that was the design of a mechanism to improve scalp–optode contact in fDCS measurements.

Across the four recorded fDCS channels, the detected signal intensity increased from approximately 1 kHz in the absence of pressure to around 6.5 kHz under full pressure, representing a threefold enhancement sustained over 70 seconds. Phase 2 exhibited limited variability, while Phase 3 showed a moderate increase in spread, suggesting consistent improvement across channels even in the presence of dense, dark hair.

Although typical recordings under optimal conditions can reach intensities of up to 20 kHz, several factors may have contributed to the lower values observed in this test. First, the subject's dense and dark hair may have introduced greater photon attenuation than is expected in 4–5-month-old infants. Second, the pressure mechanism includes a safety ring designed to limit compression force, calibrated specifically for infant scalp compliance. This mechanical constraint may have prevented the application of sufficient pressure to overcome the higher density of adult hair. Third, the probe was not directly printed in black but was coated post-printing; this surface treatment could have introduced additional scattering or absorption artefacts, potentially affecting signal transmission.

Taken together, these elements may partially explain the why the signal stabilizes around 6.5 kHz. While the results suggest that the pressure mechanism performs as intended in maintaining stable contact, further testing under infant-specific conditions will be required to confirm its effectiveness in the target population. Further analysis is required as optode–scalp contact could not be ensured in this single test.

Finally, the full setup, including lateral probes (2), occipital probe (1) pressure mechanisms, wiring and optical fibers, EEG cap, and simulated fNIRS and fDCS optodes, demonstrated robust mechanical stability and anatomical compatibility during proof-of-concept trials. No bending nor overlapping was observed in the optical fibres, reducing the risk of signal degradation or damage. Moreover, source–detector separations conformed to the intended specifications for both fDCS (1.8 cm) and fNIRS (2.9 cm), ensuring compatibility with physiological depth targets. The spatial arrangement of optodes successfully preserved multimodal overlap across the four target regions. Furthermore, the pressure mechanism components remained securely in place despite their vertical orientation, suggesting that the system provides sufficient mechanical retention for use in real recordings with infants.

4.1 Limitations

Despite the advances demonstrated in the setup developed in this Bachelor's Thesis, several limitations should be noted. First, the probe bodies were painted rather than directly printed entirely in black resin, due to recurring failures of the available black elastic resin in the SLA printer of the laboratory. Therefore, the probe was printed in transparent material and subsequently coated in black, an approach that may introduce variability in light absorption and can contaminate signals compromising optical measurements. Moreover, the material proved fragile during assembly; inserting the EEG electrodes into the holes caused the probe structures to break.

Second, the test was conducted on a single adult subject, while the true fit, comfort and data integrity in 4–5-month-old infants remain unverified and require pilot studies.

Third, although the system has been validated in an adult for concurrent fDCS recording, the fNIRS optodes have not yet been tested *in situ*. This leaves open the possibility of crosstalk, that is, unwanted interference where one modality's signals contaminate another, when all three techniques operate simultaneously.

Finally, the bottom part of the optode pressure mechanism should be reduced by 0.5 mm, as the optode fits loosely and shows slight movement when inserted.

4.2 Future work

Building on the limitations identified above, several next steps for future development are planned. First, probe bodies should be directly printed in black, eliminating the need for post-print coating and ensuring a uniform, non-reflective finish. Moreover, the mechanical fragility observed during assembly highlights the need for a more resistant flexible material. Future iterations of the probe should explore alternative materials or fabrication methods that maintain flexibility while significantly improving structural integrity and robustness, especially during the insertion of EEG electrodes.

Second, dedicated pilot studies with 4–5-month-old infants with a range of different hair densities will validate *in vivo* fit, comfort and signal quality across EEG, fNIRS and fDCS channels. In addition, the existing pipeline involving the created python script and the Autodesk Fusion models, can be extrapolated to other studies that target different age groups or cortical regions, by adapting the code and corresponding 3D probe designs.

Third, the bottom part of the mechanical pressure mechanism has to be slightly enlarged to prevent detector movement.

Finally, a detailed regulatory and ethics strategy must be developed. Even though the study is conducted outside a hospital setting, any setup placed on an infant's head requires evaluation by the Ethics Committee to ensure participant safety, proper informed consent procedures and compliance with data protection regulations. Therefore, a submission outlining setup specifications, biocompatibility evidence, risk-benefit analyses and monitoring plans should be prepared.

5. CONCLUSION

This Bachelor's Thesis aimed to contribute to the development of a multimodal neuroimaging system to study early language processing in 4–5-month-old infants. Three main objectives were addressed: enhancing data usability from previous recordings, designing a pressure mechanism to improve optode–scalp coupling in fDCS, and creating a sensor distribution that integrates EEG, fNIRS and fDCS while respecting anatomical and technical constraints.

First, re-analysis of visual recordings allowed the recovery of 23 participants previously discarded, with 18 ultimately meeting quality standards, demonstrating that refined analysis can increase data usability in infant research.

Second, the thread-actuated pressure mechanism was designed for improving fDCS signal intensity and tested in an adult, showing that the concept works under hair-dense conditions. However, refinements and *in-vivo* trials with infants are still required to validate comfort and safety in the target population.

Third, a Python to Autodesk Fusion 360 workflow allowed the development of a final layout that maintains fNIRS (2.9 cm) and fDCS (1.8 cm) source–detector separations within target ranges, guarantees multimodal overlap for fDCS and fNIRS measurements, routes fibres without bending and fits within the 40–43 cm infant head circumference.

Together, these outcomes represent a significant step toward a reliable multimodal neuroimaging system for early infancy. Although specific refinements are still required for each component, the progress achieved in this thesis provides the essential tools needed to finalise a robust multimodal setup for infant research.

Author Declarations

The Author, Silvia Valls Santafé declares that all cited sources have been read and are sufficiently contrasted, to be deemed reliable.

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SUPPORTING INFORMATION (SI)

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1. Extended methodology

SI - 1.1 Stimulus and workflow design

In addition to addressing the lack of spatial resolution present in study [3], the new study also aims to overcome several other methodological limitations. Firstly, the original study [3] only computed the power in the data analysis. In this new study, apart from measuring power, one of its main objectives is to assess entrainment, that is the synchronization of neural oscillations to external rhythmic stimuli, in monolingual and bilingual infants. To achieve this, the phase log in value will be computed, as it is a measure of entrainment. To be able to compute these measurements, the auditory stimulus will be recorded.

This study will focus on only two languages, significantly reducing the number of languages compared to [3] (from three to two). This decision was made because [3] reported a low number of valid trials per language condition.

Moreover, in the original study [3], 2 rhythmic categories were included (syllable-timed and stress-timed), but in this new one, only 1 rhythmic category will be included. This decision is based on the findings of [3], where the largest differences between monolingual and bilingual infants were observed in the native vs. Italian condition, both belonging to the same rhythmic category. In contrast, responses to the native vs. German condition (different rhythmic categories) were more similar. For these reasons, the stimuli will include recordings from Catalan speakers and Greek speakers, speaking their native languages: Catalan as the native one and Greek as a language from the same rhythmic category (syllable-timed) but foreign. The choice of Catalan as the native language instead of Spanish is based on the participant distribution database of Center for Brain and Cognition. Among the infants evaluated since 2020, 904 are exposed to Catalan (329 monolinguals and 575 bilinguals), compared to 833 exposed to Spanish (370 monolinguals and 463 bilinguals). Given the higher number of Catalan listeners infants, using Catalan as the stimulus language increases the likelihood of consistent language exposure across participants.

The design of the temporal structure of the stimuli is carefully optimized to balance behavioural constraints in infant participants. Silent intervals between stimuli must remain under 22 seconds, since in a study [ongoing study] of the SAP group measured via an accelerometer in the occipital region, demonstrated that in the absence of stimulation, motion artifacts increase. However, these silences cannot be too short either, since a minimum duration is necessary to allow the hemodynamic response to return to baseline, ensuring the validity of successive measures [15].

Furthermore, given that EEG has a rapid temporal response while fNIRS and fDCS exhibit slower responses, the visual stimuli design must balance these temporal characteristics to achieve optimal and reliable measurements across each modality.

SI -1.2 Conceptual design analysis

SI - 1.2.1 Screw gear optode pressure mechanism

This system combined a screw with an internal gear to control the vertical position of the zipper. The user had to rotate the screw from the outside of the casing, which turned a gear inside. The gear moved the arm up or down with high precision, allowing fine pressure adjustment on the scalp. A hole on the lateral side of the casing allowed the optode to enter the capsule, where it was then pushed downward toward the scalp by the internal zipper mechanism. The entire adjustment system was fully enclosed inside a capsule, with only the screw accessible. This ensured safe use in infants, preventing contact with moving parts and maintaining a clean and a soft interface with the head.

SI - 1.2.2 Dual screw based optode pressure mechanism

This design used two screws, one horizontal and one vertical, to control both the position and pressure of the optode. The horizontal screw, accessible from the outside of the casing, adjusted the position of the vertical screw, which in turn pressed the optode down toward the scalp. This system enabled precise adjustment, helping to improve signal quality while adapting to different head shapes and hair conditions.

SI - 1.2.3 Magnetic optode pressure mechanism

This system used magnets to press the optode against the scalp and improve contact through the hair. It was designed to be mounted on a cap, with one magnet placed above the upper support and another beneath the lower support, so that their mutual attraction held the optode securely in place. Magnets with different strengths could be used depending on the area of the head and the amount of hair. This allowed the system to adapt easily and apply the right pressure without using complex parts, keeping the setup simple and safe for infants.

SI - 1.2.4 Vertical pressure applicator with internal piston

This design explored a vertically located piston system intended to apply adjustable pressure on the optode. The system consisted of a bottom casing with two openings: one for inserting the optode body and another for its tip, aligned with the contact surface. Inside the bottom casing, a support mechanism had a central piston designed to apply vertical pressure when actuated through an adjustable top cover. The mechanism functioned by pressing the top cover, which in turn pushed the internal piston downward, transmitting force to the optode to improve contact. The concept aimed to offer a mechanical system without external components, allowing easy adjustment.

SI - 1.3 Exploration of physical prototypes

After the rejection of the initial 3D designs, explained in section *3.2 Preliminary conceptual analysis of the pressure mechanisms* the development process shifted toward rapid conceptual validation through hands-on prototyping. Given the time consumption of Autodesk Fusion modelling, especially for designs that might later be discarded due to basic feasibility issues, an engineering approach was adopted to accelerate iteration.

These models served to physically test mechanical concepts under realistic constraints, such as space or stability on curved surfaces. Additionally, the materials used in these prototypes were not those intended for final production; each model was assembled using accessible household materials and improvised components, prioritizing speed and functionality over realism.

SI - 1.3.1 Lever-based pressure mechanism prototype

The first prototype was a lever-based system designed to apply controlled pressure through a mechanical lever. The system integrated several components: a compression lever, a lateral locking tab integrated into a spring, a pressure rod connected to a circular support base and an infant-friendly cover. All components were housed within a cylindrical casing featuring a lateral opening to insert the optode and route its fibre.

Pressing the lever lifts the locking tab, allowing the pressure rod to move. With the lever pressed, the tab can be positioned into the chosen level (1, 2 or 3). Level 1 yields the smallest downward travel of the rod, while level 3 produces the greatest travel and strongest pressure. Releasing the lever causes the tab to engage that level and lock the rod place. Moreover, a side opening allows the optode to enter without obstruction.

SI - 1.3.2 Modular compression system

The second prototype was a compression system composed of only two main parts: an adjustment silicone component with four predefined height levels and a rigid circular support base with a lateral opening for optode insertion.

The optode was pressed toward the scalp by the downward compression of the silicone structure to the plastic part. This design was lightweight, easy to assemble and its minimal part count simplified both fabrication and integration into existing cap systems.

SI - 1.3.3 Spring-loaded system

The third prototype featured a system structured around two main components. First, an upper casing with an integrated screw mechanism. The interior of the casing included a threaded part, allowing the screw to rotate and advance downward. The optode had to be integrated laterally through an opening in the casing and guided downward toward the scalp by the action of the screw. The base component was divided into two circular elements, one positioned above the cap and the other below, providing a stable anchoring point for the casing.

2. Supplementary Figures

2.1 Visual study

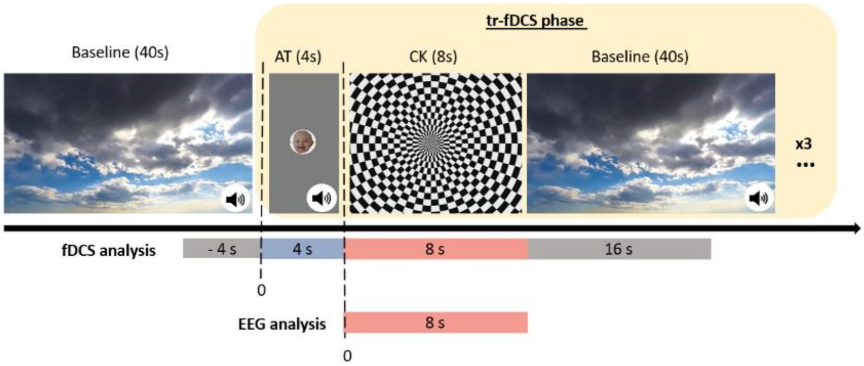


Figure SI- 1. Experimental protocol and visual stimuli. [Article in preparation]. Only Phase 1 is shown, as it is the one analysed in this BT. It consists of three trials, each including 4 seconds of AT, 8 seconds of the CK pattern and 40 seconds of BL. The analysis windows for fDCS and EEG are indicated below the timeline.

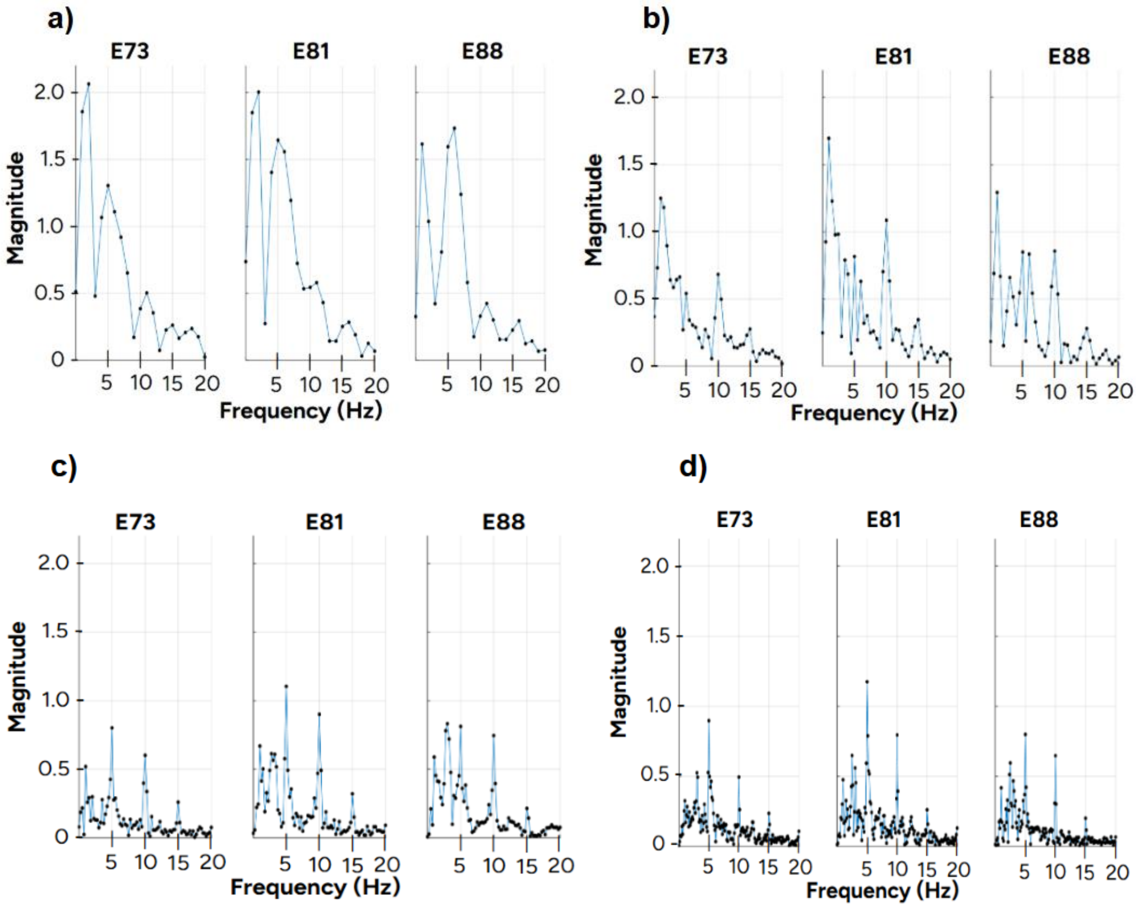


Figure SI- 2. Fast-Fourier Transform (FFT) spectra at electrodes E73, E81 and E88 for a 5 Hz visual stimulation, displayed across four different recording durations: a) 1 second, b) 2 seconds, c) 4 seconds and d) 8 seconds.

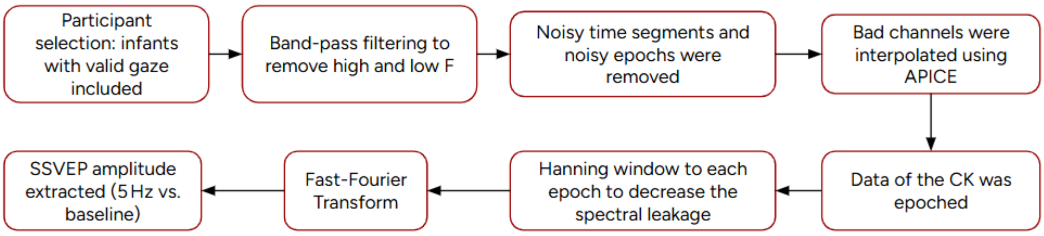


Figure SI- 3. EEG preprocessing and analysis pipeline. Schematic representation of the signal processing steps applied to EEG data from infants with valid gaze.

2.2 Development of an optode pressure mechanism

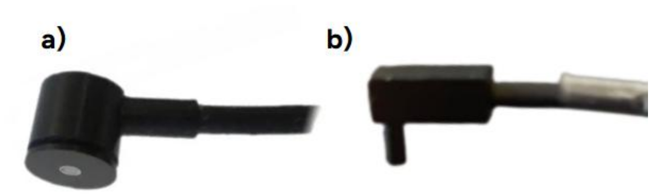


Figure SI- 4. Comparison of optode shapes used in fNIRS (a) and fDCS (b) systems. (a) Cylindrical optode with flat contact surface, commonly used for broader scalp contact. (b) Rectangular optode with a protruding contact tip, designed for targeted signal application or acquisition.

Figure SI- 5. Visual comparison of connection systems used in commercial fNIRS devices. The figure presents representative images of the mechanisms from four major manufacturers (NIRx, LUMO, CortiVision, and Artinis), organized in four columns: the first column lists the system names, the second shows the optode designs, the third displays the caps used for sensor placement and the fourth provides additional views for detailed examination. This comparison supports the critical analysis of existing solutions discussed in section *2.4.1 Analysis of reference systems for optode fixation.* All images were obtained directly from the physical equipment available in ICFO and are not sourced from online websites.

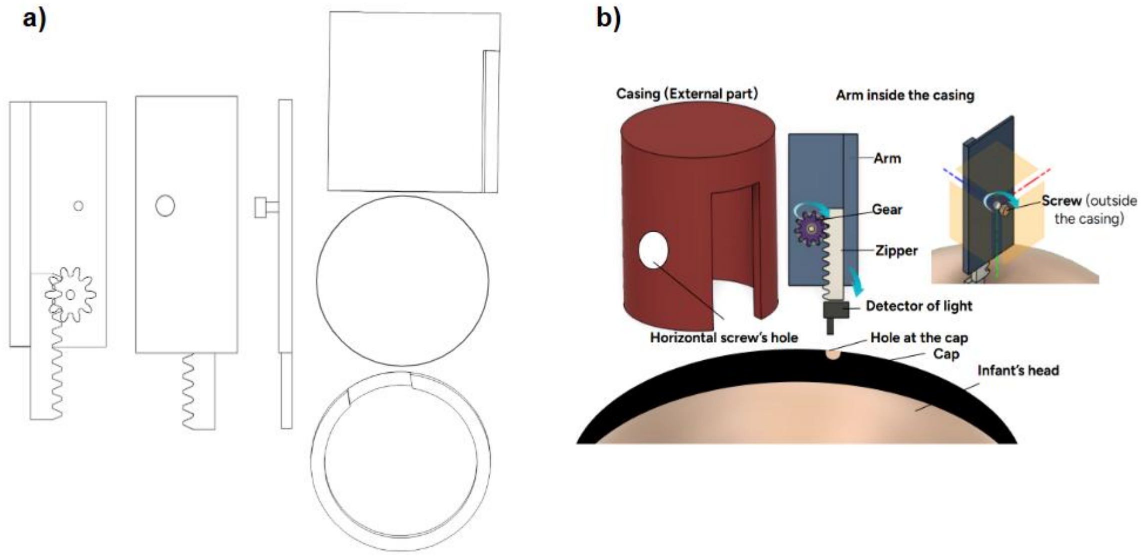


Figure SI- 6. Design of the adjustable screw gear mechanism. a) Sketch of the individual components b) 3D design generated in Autodesk Fusion 360 of the mechanism.

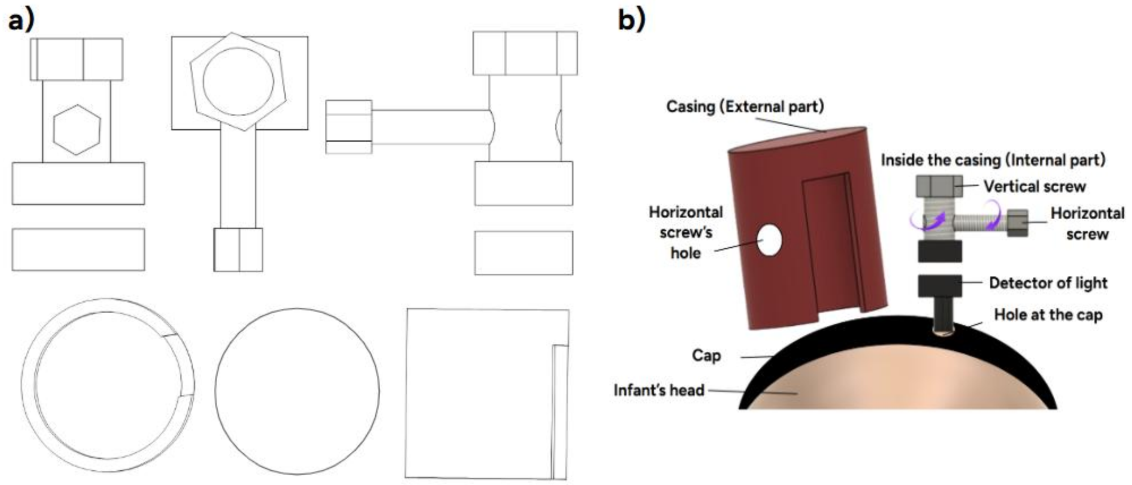


Figure SI- 7. Design of the adjustable dual screw mechanism. a) Sketch of the individual components. b) 3D design generated in Autodesk Fusion 360 of the mechanism.

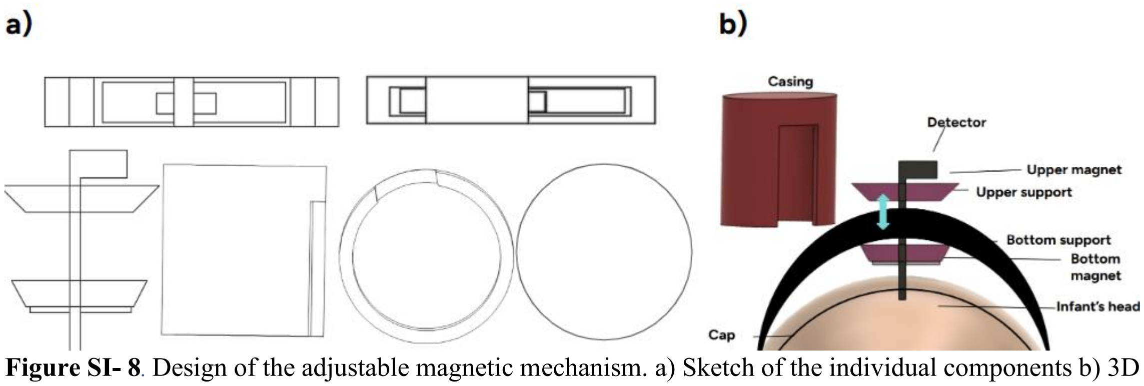


Figure SI- 8. Design of the adjustable magnetic mechanism. a) Sketch of the individual components b) 3D design generated in Autodesk Fusion 360 of the mechanism.

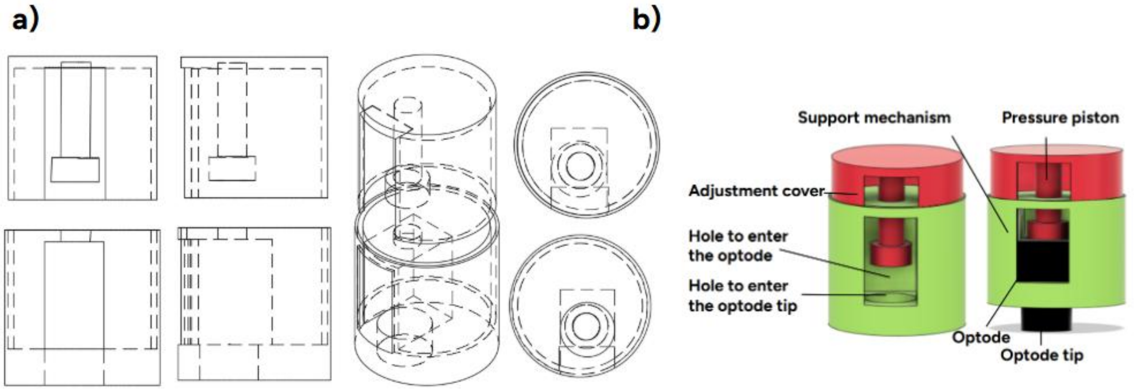


Figure SI- 9. Design of the adjustable vertical pressure applicator. a) Sketch of the individual components b) 3D design generated in Autodesk Fusion 360 of the mechanism.

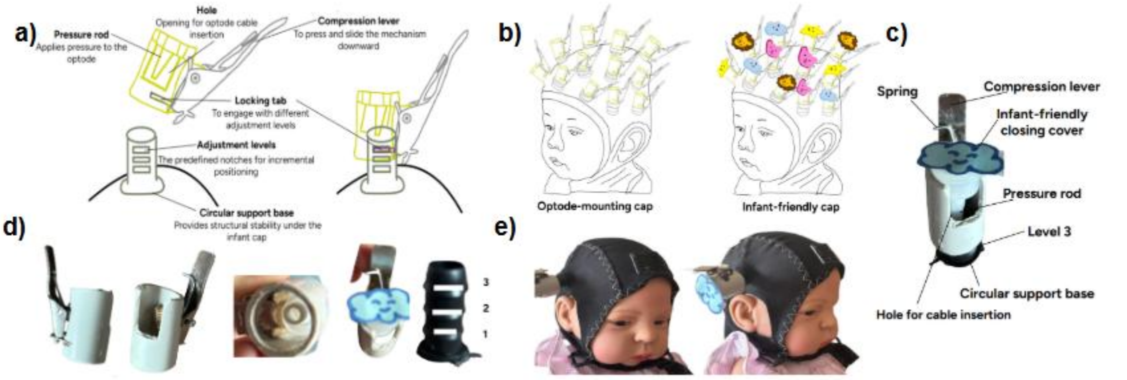


Figure SI- 10. Lever-based pressure mechanism prototype. a) Diagram of the mechanism with adjustable pressure levels and a locking system for optode positioning. b) Initial sketch of the desired infant-friendly cap. c) Annotated view of the mechanism highlighting its components. d) Mechanical prototype parts. e) Final system fitted on a baby mannequin using the soft, infant-friendly cap design.

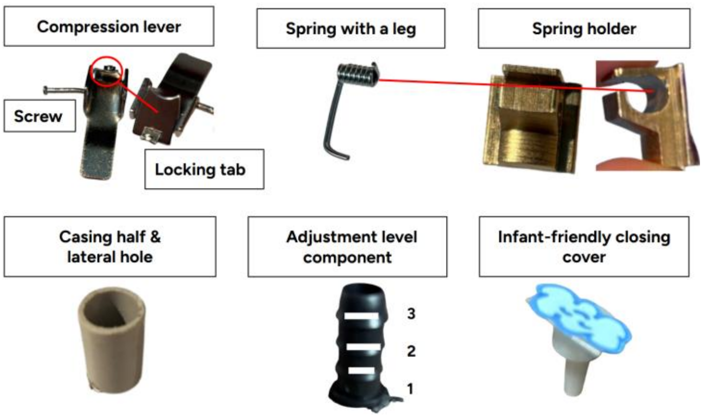


Figure SI- 11. Different components required to assemble the lever-based mechanism.

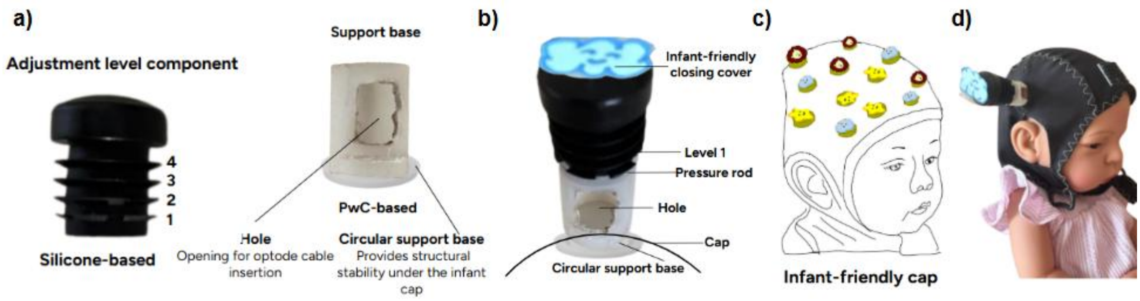


Figure SI- 12. Modular compression system prototype. a) Main components of the system. b) Assembled mechanism in the lowest level. c) Initial sketch of the aimed infant-friendly cap. e) Final implementation on an infant-mannequin demonstrating proper fit and secure attachment under the cap.

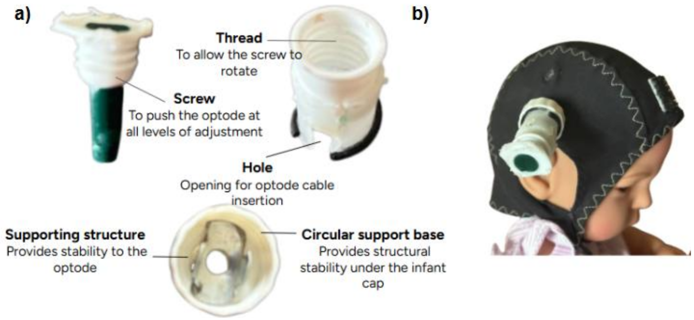


Figure SI- 13. Spring-loaded compression mechanism. a) Components of the spring-loaded system. b) Final prototype mounted on an infant mannequin, demonstrating proper alignment and secure attachment through the cap.

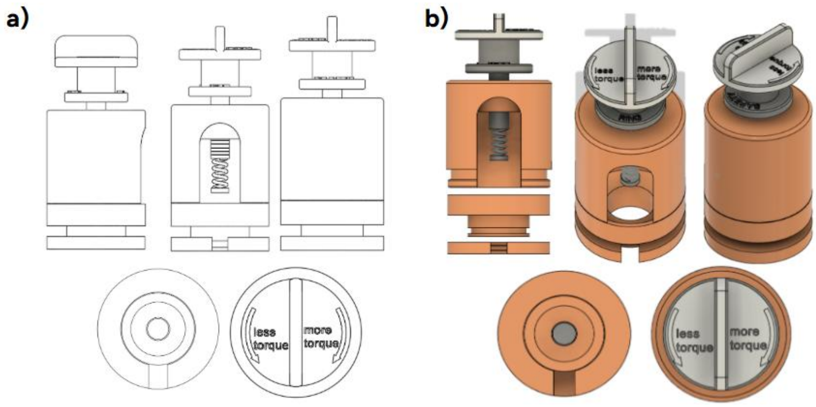


Figure SI- 14. Different views of adjustable spring-loaded optode pressure mechanism designed for cap integration. a) Sketch of the individual components b) 3D design generated in Fusion 360 of the mechanism.

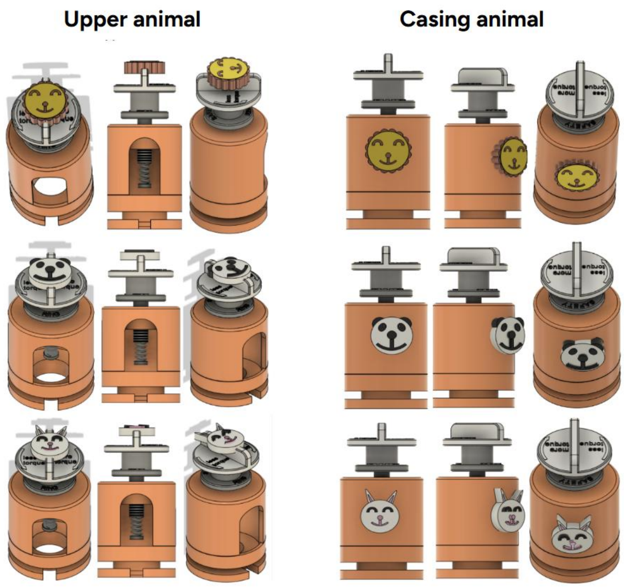


Figure SI- 15. Different views of rejected designs of the infant-friendly support to the spring-loaded mechanism. Two design strategies were explored to incorporate decorative animal elements on the mechanism but were ultimately rejected in favor of a lighter alternative.

2.3 Development of the multimodal probe

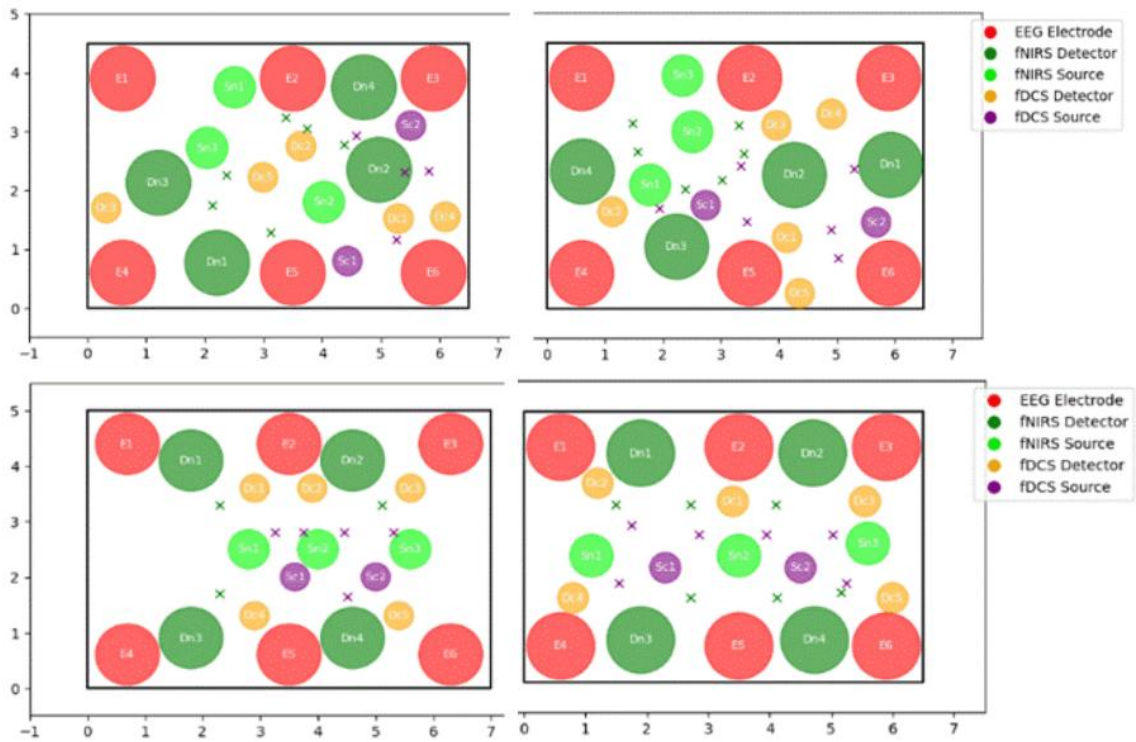


Figure SI- 16. Initial sensor distribution layouts generated with Python. These plots correspond to the first configurations created during the early design phase using 2D code-based prototyping. Each panel represents a different candidate layout for the multimodal probe, exploring several spatial arrangements of sensors. This first version of the code served as the starting point for the design process and was later refined to improve visualization and functionality. These initial layouts provided the foundation for subsequent design iterations.

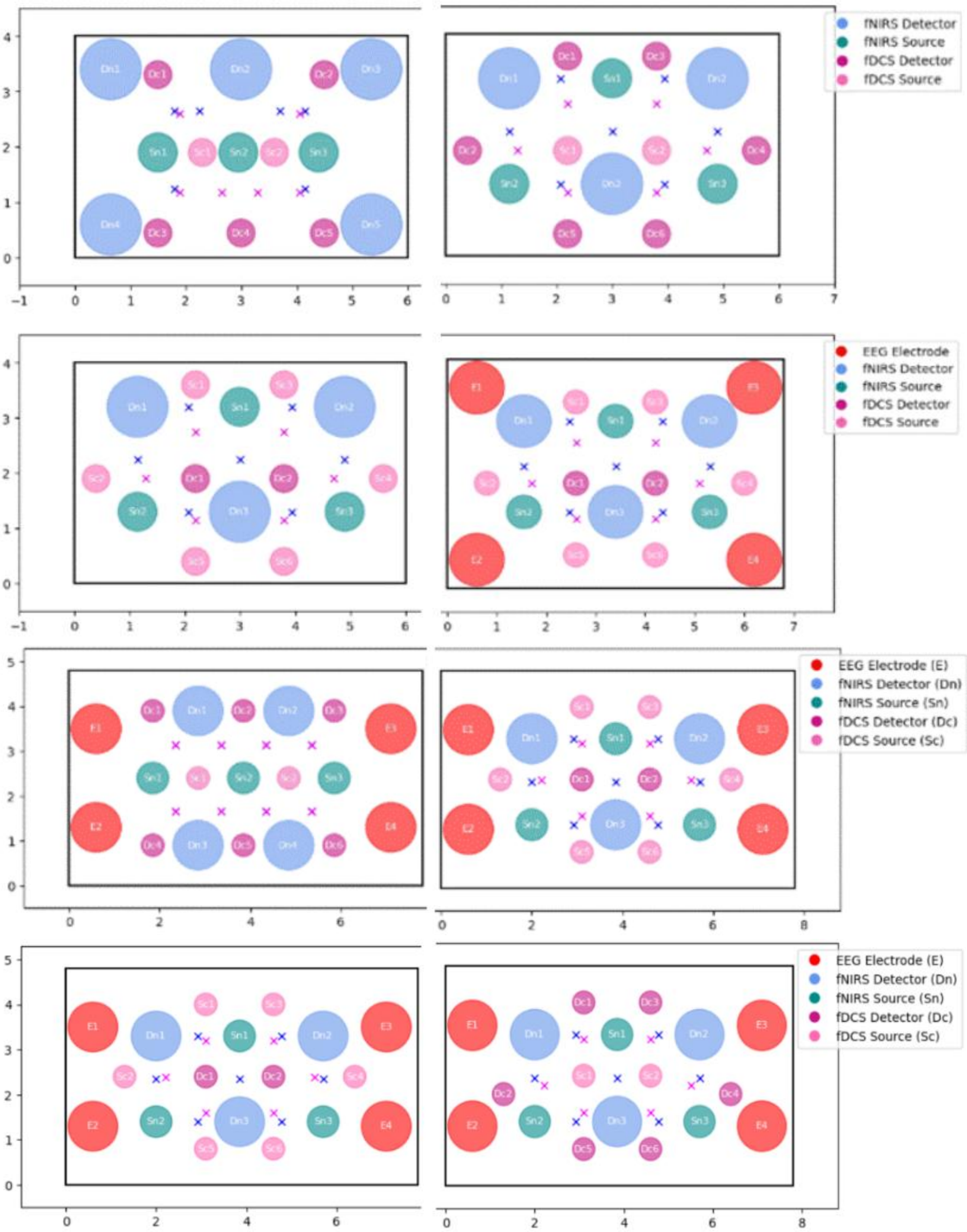


Figure SI- 17. Sensor distribution layouts generated using the refined version of the Python code. These plots correspond to a selection of candidate configurations tested during the intermediate design phase. The updated code allowed for improved visualization and flexible parameter control, enabling more systematic exploration of sensor placement. While many other models were generated and evaluated during this stage, the ones shown here represent the most relevant distributions that guided the final design decisions.

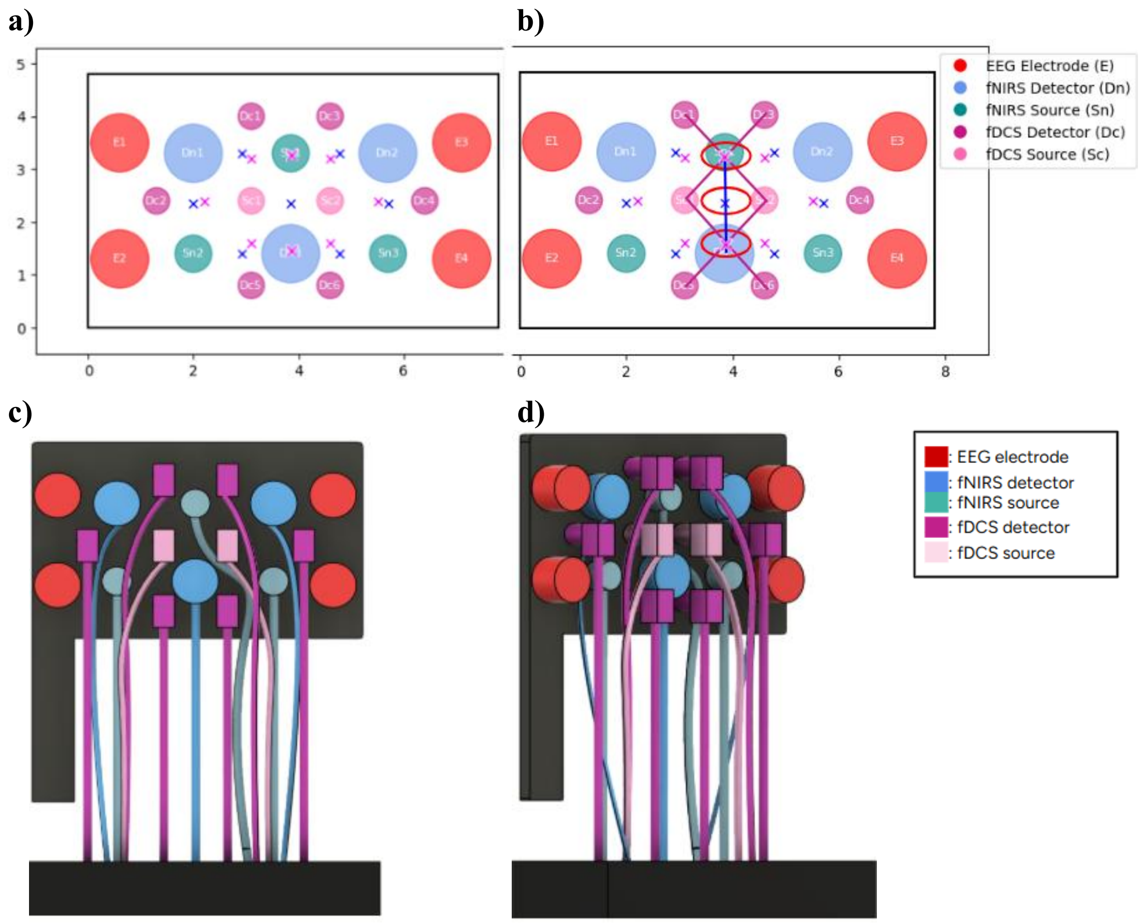


Figure SI- 18. Spatial configuration of EEG electrodes, fNIRS optodes and fDCS optodes of the second approved design configuration. Panel (a) displays the full distribution of sensors on a rectangular probe surface. Panel (b) illustrates the challenge in the measurement regions for each optical modality, with blue and red ellipses denoting areas exclusively covered by fNIRS and fDCS, respectively. Panels (c) and (d) show the corresponding 3D model developed in Autodesk Fusion 360, providing front and left-front views, respectively. This model allowed evaluate fibre (in fDCS) and wire (in fNIRS) routing feasibility.

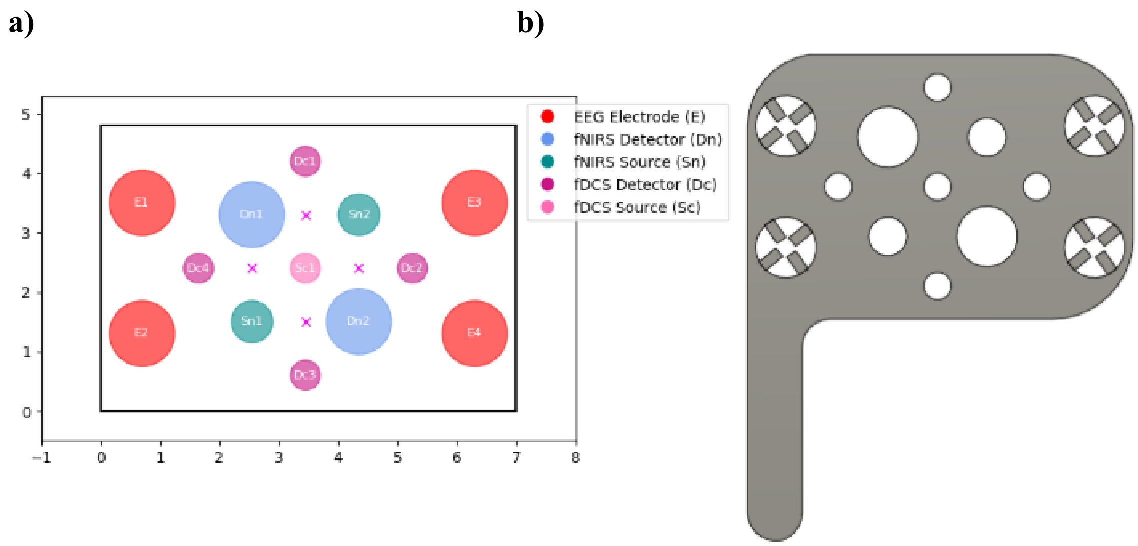
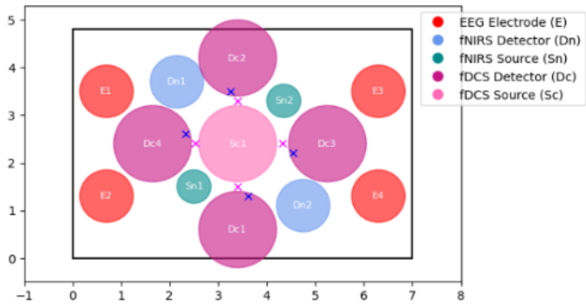
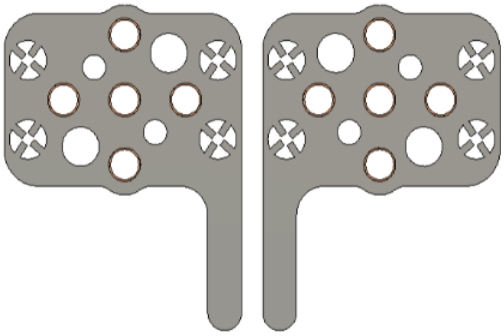


Figure SI- 19. Second validated lateral multimodal layout. (a) Sensor distribution plot generated with Python including EEG electrodes, fNIRS optodes and fDCS optodes, arranged within a 4.8 × 7.0 cm rectangular probe surface, displaying the four measurement areas with complete overlap between fNIRS and fDCS modalities. (b) Corresponding 3D model of the structure developed in Autodesk Fusion 360, designed with circular apertures as provisional placeholders for optode placement, prior to integration of the final fixation mechanism.

a)



b)



c)

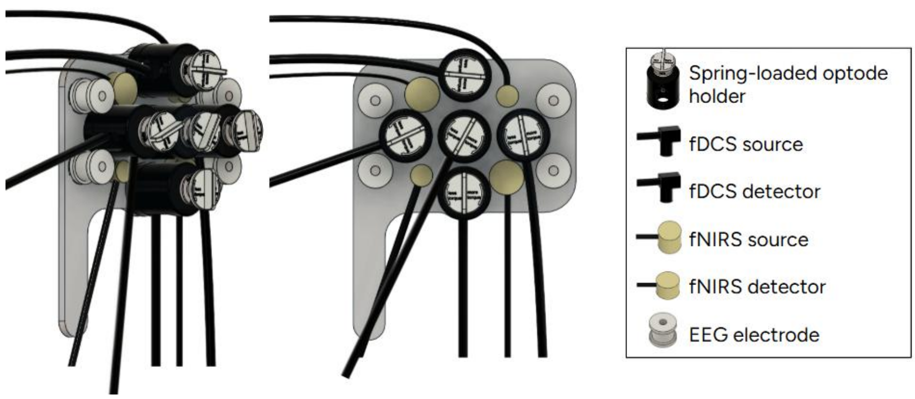
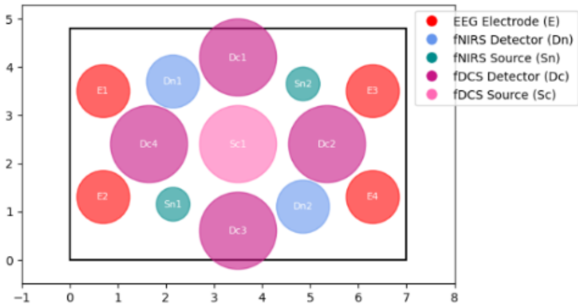
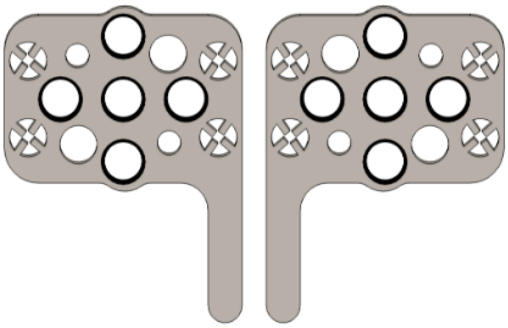


Figure SI- 20. Design and configuration of the first printed lateral multimodal probe. Panel (a) presents the sensor layout generated with Python, illustrating the spatial arrangement within a 4.8 × 7.0 cm rectangular probe surface. Panel (b) displays the 3D design of the structure for both left and right hemispheres, with circular openings for sensor placement. Panel (c) illustrates the complete 3D assembly including all sensors, wires and fibres, used to verify that all components fit properly and that there is enough space for correct routing. The models in panels (b) and (c) were created using Autodesk Fusion 360.

a)



b)



c)

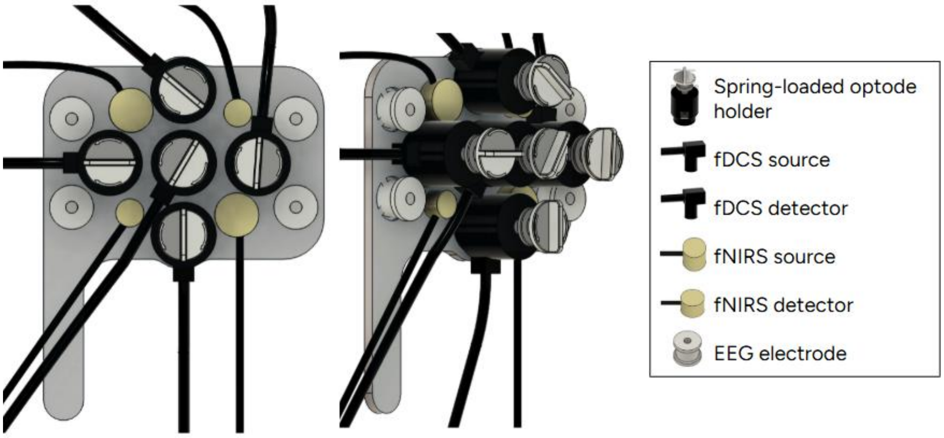
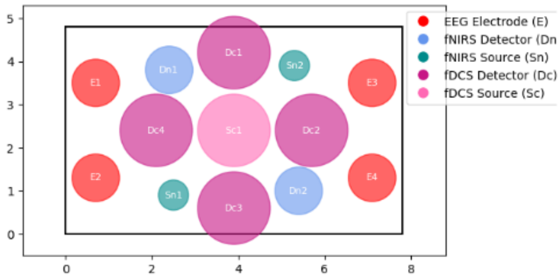
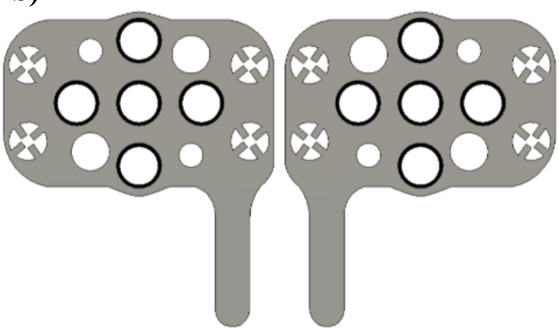


Figure SI- 21. Design and configuration of the second printed lateral multimodal probe. Panel (a) presents the sensor layout generated with Python, illustrating the spatial arrangement within a 4.8 × 7.0 cm rectangular probe surface. Panel (b) displays the 3D design of the structure for both left and right hemispheres, with circular openings for sensor placement. Panel (c) illustrates the complete 3D assembly including all sensors, wires and fibres, used to verify that all components fit properly and that there is enough space for correct routing. The models in panels (b) and (c) were created using Autodesk Fusion 360.

a) **b)**





c)

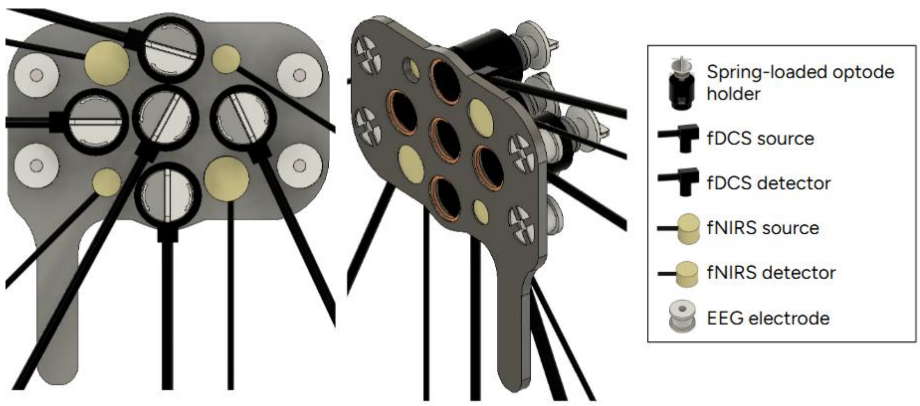


Figure SI- 22. Design and configuration of the final lateral multimodal probe. Panel (a) presents the sensor layout generated with Python, illustrating the spatial arrangement within a 4.8 × 7.8 cm rectangular probe surface. Panel (b) displays the 3D design of the structure for both left and right hemispheres, with circular openings for sensor placement. Panel (c) illustrates the complete 3D assembly including all sensors, wires and fibres, used to verify that all components fit properly and that there is enough space for correct routing. The models in panels (b) and (c) were created using Autodesk Fusion 360.

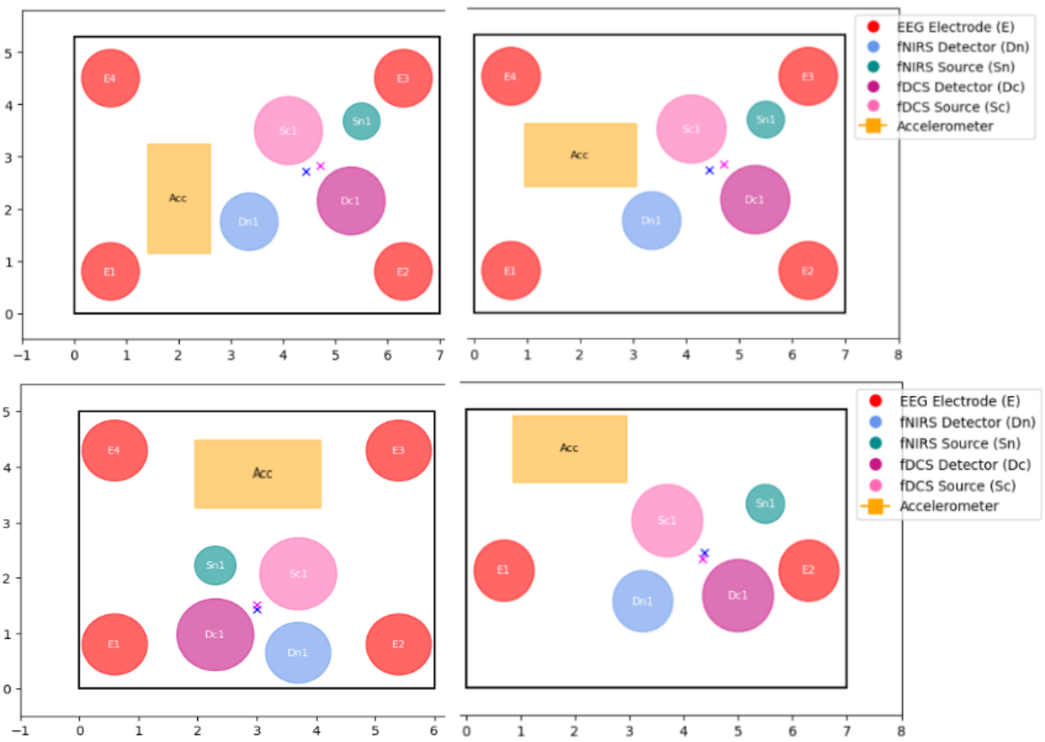


Figure SI- 23. Candidate sensor layouts for the occipital probe. This figure displays several plots that correspond to proposed configurations for the occipital probe, generated with Python. Each layout explores different spatial arrangements of the required components.

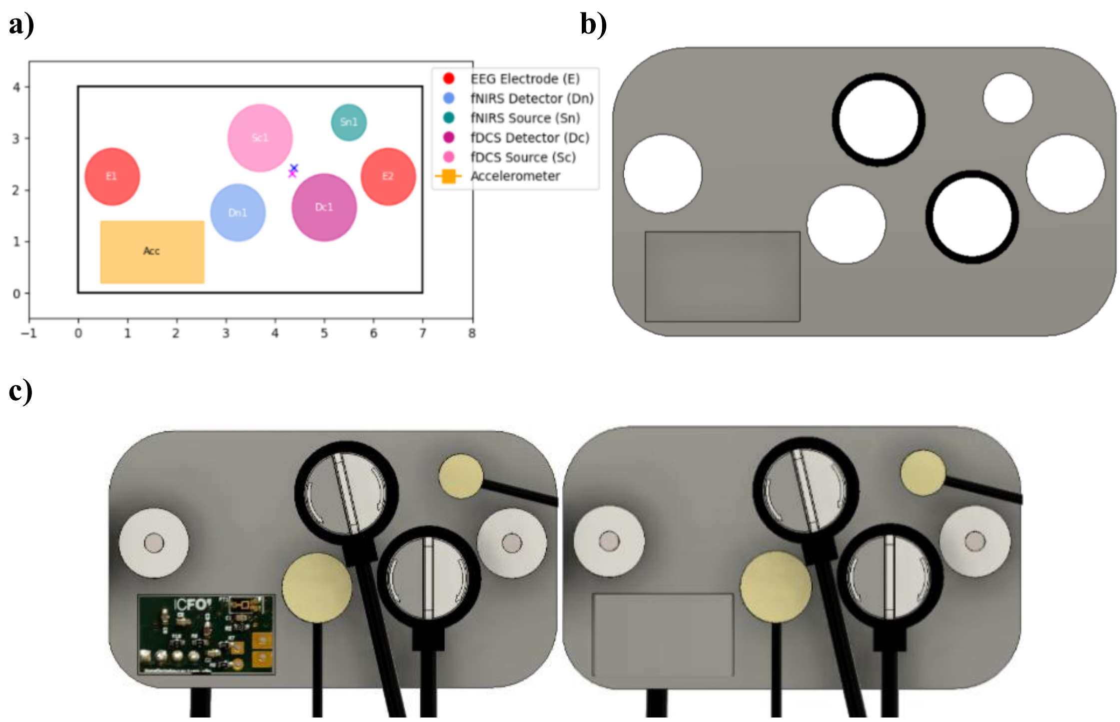


Figure SI- 24. Design and configuration of the finnal occipital multimodal probe. Panel (a) shows the layout of integrated sensors generated with Python, distributed over a rectangular surface of 4.0 × 7.0 cm. The design includes EEG electrodes, fNIRS and fDCS optodes, and a rectangular accelerometer. Panel (b) presents the 3D model with circular holes for sensor integration. Panel (c) displays the assembled design, illustrating the routing of wires and optical fibres. All 3D components were modelled using Autodesk Fusion 360.

2.4 Final setup

1. Prepare the multimodal probe.
2. Attach the probe to the EEG cap.
3. Assembly the thread actuated pressure mechanism.
4. Prepare all the required pressure mechanisms.
5. Mount all optodes and pressure mechanisms onto the probe of the EEG cap (screw with minimal pressure).
6. Fit the complete setup onto the infant.
7. Check the signal quality and adjust all screws to ensure optimal contact.

Figure SI- 25. Step by step visual guide for the assembly and placement of the multimodal neuroimaging system on an infant.

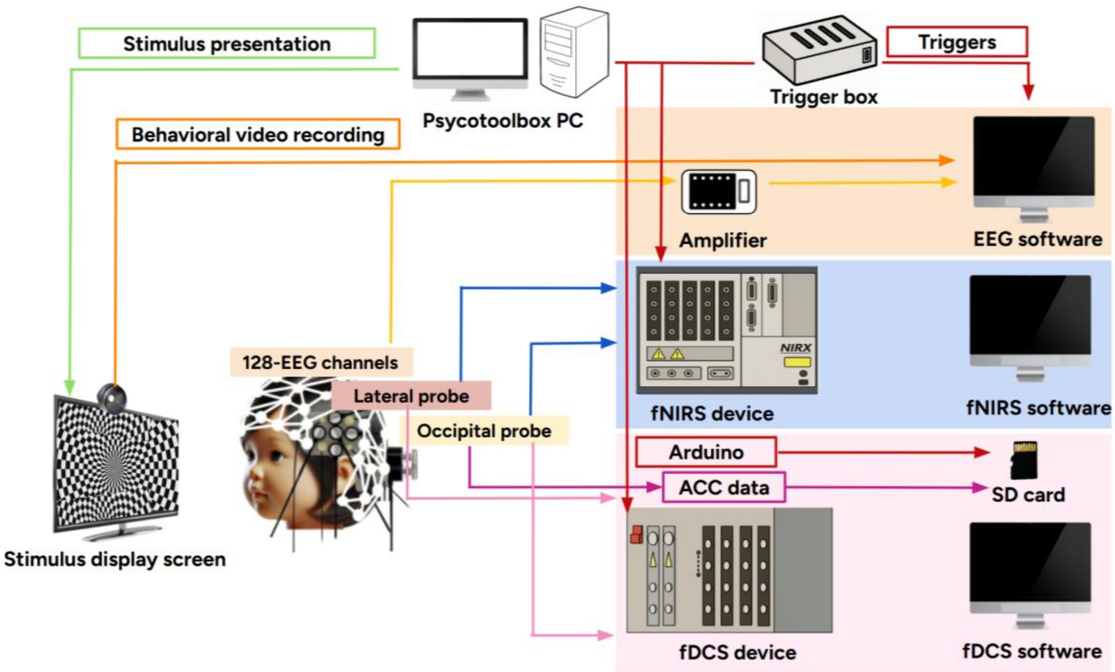


Figure SI- 26. Schematic of the multimodal experimental setup. Visual representation of the complete acquisition system integrating EEG, fNIRS and fDCS technologies using lateral and occipital probes.

3. Supplementary tables

3.1 Multimodal probe configurations

Type of sensor	Position *X* (cm)	Position *Y* (cm)
EEG Electrodes	0.60	3.50
	0.60	1.30
	7.10	3.50
	7.10	1.30
fNIRS sources	3.85	3.30
	2.00	1.40
	5.70	3.30
fNIRS detectors	2.00	3.30
	5.70	3.30
	3.85	1.40
fDCS sources	3.10	2.40
	4.60	2.40
fDCS detectors	3.10	4.00
	1.30	2.40
	4.60	4.00
	6.40	2.40
	3.10	0.80
	4.60	0.80

Table SI- 1. Spatial coordinates (in cm) of all sensors included in the first lateral approved multimodal configuration of the probe. The positions are referenced to a rectangular 2D surface (7.8 × 4.8 cm) representing the internal distribution plane of the probe casing, onto which the sensors are mounted and was used for layout simulation, spatial validation and subsequent 3D modelling. The origin of coordinates (0,0) is defined at the bottom left corner of the extended rectangular tab of the probe.

Type of sensor	Position *X* (cm)	Position *Y* (cm)
EEG Electrodes	0.70	3.50
	0.70	1.30
	6.30	3.50
	6.30	1.30
fNIRS sources	2.55	1.50
	4.35	3.30
fNIRS detectors	2.55	3.30
	4.35	1.50
fDCS source	3.45	2.40
fDCS detectors	3.45	4.20
	5.25	2.40
	3.45	0.60
	1.65	2.40

Table SI- 2. Spatial coordinates (in cm) of all sensors included in the second lateral approved multimodal configuration of the probe. The positions are referenced to a rectangular 2D surface (7 × 4.8 cm) representing the internal distribution plane of the probe casing, onto which the sensors are mounted and was used for layout simulation, spatial validation and subsequent 3D modelling. The origin of coordinates (0,0) is defined at the bottom left corner of the extended rectangular tab of the probe.

Type of sensor	**Position** *X* **(cm)**	**Position** *Y* **(cm)**
EEG Electrodes	0.70	3.50
	0.70	1.30
	6.30	3.50
	6.30	1.30
fNIRS sources	2.50	1.50
	4.35	3.30
fNIRS detectors	2.15	3.70
	4.75	1.10
fDCS source	3.40	2.40
fDCS detectors	3.40	0.60
	3.40	4.20
	5.25	2.40
	1.65	2.40

Table SI- 3. Spatial coordinates (in cm) of all sensors included in the third lateral approved multimodal configuration of the probe. The positions are referenced to a rectangular 2D surface (7 × 4.8 cm) representing the internal distribution plane of the probe casing, onto which the sensors are mounted and was used for layout simulation, spatial validation and subsequent 3D modelling and 3D printing. The origin of coordinates (0,0) is defined at the bottom left corner of the extended rectangular tab of the probe.

Type of sensor	**Position** *X* **(cm)**	**Position** *Y* **(cm)**
EEG Electrodes	0.70	3.50
	0.70	1.30
	6.30	3.50
	6.30	1.30
fNIRS sources	2.15	1.15
	4.85	3.65
fNIRS detectors	21.8	3.70
	4.82	1.10
fDCS source	3.50	2.40
fDCS detectors	3.50	4.20
	5.35	2.40
	3.50	0.60
	1.65	2.40

Table SI- 4. Spatial coordinates (in cm) of all sensors included in the fourth lateral multimodal configuration of the probe. The positions are referenced to a rectangular 2D surface (7 × 4.8 cm) representing the internal distribution plane of the probe casing, onto which the sensors are mounted and was used for layout simulation, spatial validation and subsequent 3D modelling and 3D printing. The origin of coordinates (0,0) is defined at the bottom left corner of the extended rectangular tab of the probe.

Type of sensor	Position X (cm)	Position Y (cm)
EEG Electrodes	0.70	3.50
	0.70	1.30
	7.10	3.50
	7.10	1.30
fNIRS source	2.40	0.90
	5.30	3.90
fNIRS detectors	2.40	3.80
	5.30	1.00
fDCS source	3.90	2.40
fDCS detectors	3.90	4.20
	5.70	2.40
	3.90	0.60
	2.10	2.40

Table SI- 5. Spatial coordinates (in cm) of all sensors included in the final lateral multimodal configuration of the probe. The positions are referenced to a rectangular 2D surface (7.8 × 4.8 cm) representing the internal distribution plane of the probe casing, onto which the sensors are mounted and was used for layout simulation, spatial validation and subsequent 3D modelling and 3D printing. The origin of coordinates (0,0) is defined at the bottom left corner of the extended rectangular tab of the probe.

Type of sensor	Position X (cm)	Position Y (cm)
EEG Electrodes	0.70	2.25
	6.30	2.25
fNIRS source	5.50	3.30
fNIRS detector	3.25	1.55
fDCS source	3.70	3.00
fDCS detector	5.00	1.65
Accelerometer	0.45	0.20

Table SI- 6. Spatial coordinates (in cm) of all sensors included in the occipital multimodal configuration of the probe. The positions are referenced to a rectangular 2D surface (7 × 4 cm) representing the internal distribution plane of the probe casing, onto which the sensors are mounted and was used for layout simulation, spatial validation and subsequent 3D modelling and 3D printing. The origin of coordinates (0,0) is defined at the bottom left corner of the extended rectangular tab of the probe.

3.2 Parameters of the 3D printing process

Parameters	Screw	Casing & Bottom part
Material	Tough 2000	Black V4
Volume	8.02 mL	16.38 mL
Layers	609	854
Time	3 hours 29 minutes	8 hours 30 minutes

Table SI- 7. Printing parameters defined in PreForm software for the first printed version of the optode pressure mechanism, showing the configuration used for each component.

Parameters	First printed probe	Second printed probe	Final lateral probe	Occipital probe
Material	Flexible 80A	Flexible 80A	Flexible 80A	Flexible 80A
Volume	47.3 mL	47 mL	52 mL	37.9 mL
Probe thickness	3 mm	3 mm	3 mm	4 mm
Layers	1350	1350	1350	1458
Time	10 hours 54 minutes	10 hours 46 minutes	12 hours 1 minute	12 hours 56 minutes

Table SI- 8. Printing parameters defined in PreForm software for the printed versions of probes.